

AOS-CX 10.06 ACLs and Classifier Policies Guide

6200 Switch Series



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This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing Aruba switches on a network.

Applicable products

This document applies to the following products:

Aruba 6200 Switch Series (JL724A, JL725A, JL726A, JL727A, JL728A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in **Support and other resources**.

Command syntax notation conventions

Convention	Usage
<code>example-text</code>	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([]).
example-text	In code and screen examples, indicates text entered by a user.
Any of the following: <code><example-text></code> <code><example-text></code> <i>example-text</i> <i>example-text</i>	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: - For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value. - For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
{ }	Braces. Indicates that at least one of the enclosed items is required.

Table Continued

Convention	Usage
[]	Brackets. Indicates that the enclosed item or items are optional.
... or	Ellipsis:
...	- In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. - In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment. The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term `switch`, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch(CONTEXT-NAME)#
```

Indicates the configuration context for a feature. For example:

```
switch(config-if) #
```

Identifies the interface context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100) #
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>) #
```

Where `<VLAN-ID>` is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format:

```
member/slot/port
```

On the 6200 Switch Series

- *member*: Member number of the switch in a Virtual Switching Framework (VSF) stack. Range: 1 to 8. The primary switch is always member 1. If the switch is not a member of a VSF stack, then member is 1.
- *slot*: Line module number. Always 1.
- *port*: Physical number of a port on a line module.

For example, the logical interface 1/1/4 in software is associated with physical port 4 in slot 1 on member 1.

About Access Control Lists (ACLs)

Access Control Lists (ACLs) let a network administrator permit or deny passage of traffic based on network addresses, protocols, service ports, and other packet attributes. ACLs are composed of one or more Access Control Entries (called ACEs). Each ACE defines a filter criteria and an action, either **permit** or **deny**. If the traffic matches the filter criteria, the specified action is taken. The **permit** action permits the traffic to continue through the switch. The **deny** action causes the traffic to be discarded (dropped). ACEs can also log or count matching traffic.

Three ACL types are supported; IPv4, IPv6, and MAC. Each ACL type is focused on relevant frame or packet characteristics.

ACLs must be applied (using an `apply access-list` command) to take effect. ACLs can be applied to interfaces (including LAGs), VLANs, or the Control Plane.

Access Control Entries (ACEs) are listed according to priority by sequence number and processed in lowest to highest sequence number order. Each ACE attempts to match on one or more attributes of the particular traffic type. Attempted ACE matching ceases upon the first successful match. For a match to be considered successful, a packet must match all the criteria, qualifiers, and attributes of a particular ACE. Higher-numbered ACEs are only processed if no lower-numbered ACE matches. If the traffic matches no ACE in the entire ACL, the default action **deny** is taken, causing the traffic to be discarded (dropped).

When defining an ACE, if the sequence number is omitted, the ACE is auto-assigned a new sequence number that is 10 greater than the existing highest ACE sequence number. The first auto-assigned sequence number is 10. If you choose to include the ACE sequence numbers, you can use any number you like, however it is suggested that you follow the practice of entering them as 10, 20, 30, and so on. Regardless of the order in which ACEs are entered, they are stored in low-to-high sequence number order. If you enter three ACEs numbered 10, 30, 20, when creating an ACL, the ACEs are stored in the ACL as 10, 20, 30.

This simple ACL definition permits traffic passage for a particular address range and otherwise counts all nonmatching (dropped) traffic:

```
switch(config)# access-list ip network-A-udp-only
switch(config-acl-ip)# 10 permit udp any 172.16.1.0/24
switch(config-acl-ip)# 20 deny any any any count
switch(config-acl-ip)# exit
```

The main traffic characteristics that ACEs can filter on are as follows (see the full list in the ACE parameters list of the ACL commands):

- Protocol such as: ICMP, TCP, UDP
- Source and/or destination addresses (IPv4, IPv6, or MAC)
- Source and/or destination TCP/UDP ports (if applicable to the specified protocol)

A few real-world uses of ACLs are as follows:

- Restrict traffic arriving on a port, destined to a particular address or subnet by applying an ACL that matches on a destination IP address or an IP address and a mask.
- Prevent certain protocols from using a particular multicast MAC address (advertising through a port) by applying an ACL that matches on the destination MAC address.
- Prevent any IP host from accessing a particular IP port/application on a specific server by applying an ACL that matches on IP addresses and Layer 4 port.



IMPORTANT: See also **ACL and Policy hardware resource considerations**.

ACL usage tips

When using the `access-list ip` or `access-list ipv6` commands, if you enter an existing ACL-NAME, the existing ACL is modified as follows:

- Any ACE entered with a new sequence-number creates an additional ACE.
- Any ACE entered with an existing sequence-number replaces the existing ACE.

If you modify an ACL that has already been applied, it is possible that packets, blocked by the previous ACL, will briefly pass through the switch during the ACL reconfiguration.



NOTE: In a highly secure environment, it is safest to first bring down interfaces and VLANs to which an ACL has been applied before modifying the ACL. Then bring the targets of ACL application back up after completing the ACL modification. Respecting this recommendation ensures that an ACL is never partially programmed while traffic is passing through the switch.

About applying ACLs to interfaces or LAGs

You can apply an ACL to an interface or LAG to affect or control the traffic arriving on that interface or LAG (inbound) or leaving the interface or LAG (outbound), or both. A given interface or LAG supports the application of a single ACL per type, per direction. ACLs can be applied to interfaces or LAGs as follows:

- One MAC ACL inbound
- One MAC ACL outbound
- One IPv4 ACL inbound
- One IPv4 ACL outbound
- One IPv6 ACL inbound
- One IPv6 ACL outbound

Different ACLs of the same type can be used in opposite directions for MAC, IPv4, and IPv6. If you apply an ACL of a particular type, in a direction that is already in use, the switch replaces the current ACL with the new ACL.

About applying ACLs to VLANs

ACLs can be applied to VLANs in the inbound (ingress) and outbound (egress) directions.

Sequence numbering

If no sequence number is specified, the software appends new ACEs to the end of the ACL with a sequence number equal to the highest ACE currently in the list plus 10.

The sequence numbers may be resequenced using the `access-list resequence` command.

Deny ACLs

If multiple ACLs of different types are applied in the same direction, a deny ACE, whether explicit or implicit, in one ACL overrides a permit ACL in another. A deny ACE is an ACE within an ACL that uses the `deny` action keyword.

Denied ping requests

A ping request is denied when an ACL is applied on ingress or egress unless the request is explicitly permitted.

```
switch# ping 100.1.2.10
PING 100.1.2.10 (100.1.2.10) 100(128) bytes of data.
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
```

About address and port object groups

Object groups are useful for defining groups of IP addresses and Layer 4 ports for use exclusively in the two ACL-defining commands `access-list ip` and `access-list ipv6`.

Often, common groups of addresses and ports or port ranges are used repeatedly in many ACL definitions. Without address and port object groups, the same addresses and ports must be repeated in each ACL definition that uses them.

With address and port object groups, the IP addresses and ports can be defined once, using any of these commands:

- `object-group ip address`
- `object-group ipv6 address`
- `object-group port`

Once an object group is defined, the group is available for inclusion by name as the `<ADDRESS-GROUP>` and `<PORT-GROUP>` parameters in the `access-list ip` and `access-list ipv6` ACL-definition commands.

Object groups simplify the ACL definition process and help ensure consistent address and port specification across many ACLs.



NOTE: Keep in mind that it is possible to consume many hardware resource entries when using the object group commands. For example, in a typical situation, an ACE that uses object groups with 3 source addresses, 3 source L4 ports, 3 destination addresses, and 3 destination L4 ports, a total of 81 hardware entries are consumed ($3 * 3 * 3 * 3 = 81$).

ACL and ACE-related tasks

Common ACL and ACE-related tasks are as follows. Simple tasks link to the relevant command description.

Task	Command or procedure	Example
Creating an IPv4 ACL	<u>access-list ip</u>	access-list ip MY_IP_ACL 10 permit udp any 172.16.1.0/24 20 permit tcp 172.16.2.0/16 gt 1023 any 30 deny any any any count
Creating an IPv6 ACL	<u>access-list ipv6</u>	access-list ipv6 MY_IPV6_ACL 10 permit udp any 2001::1/64 20 permit tcp 2001:2011::1/64 any 30 deny any any any count
Creating a MAC ACL	<u>access-list mac</u>	access-list mac MY_MAC_ACL 10 permit any any appletalk vlan 40 20 deny any any any count
Applying an IPv6 ACL to an interface	<u>apply access-list</u>	interface 1/1/1 apply access-list ipv6 MY_IPV6_ACL in
Applying an IPv4 ACL to a LAG	<u>apply access-list</u>	interface lag 100 apply access-list ip MY_IP_ACL in
Applying an IPv4 ACL to a VLAN	<u>apply access-list</u>	vlan 10 apply access-list ip MY_IP_ACL in
Applying a MAC ACL to a VLAN	<u>apply access-list</u>	vlan 40 apply access-list mac MY_MAC_ACL in
Applying an IPv4 ACL to the Control Plane (OOBM)	<u>apply access-list control-plane</u>	apply access-list ip MY_IP_ACL control-plane vrf mgmt
Removing application of an ACL from an interface	<u>apply access-list</u>	interface 1/1/1 no apply access-list ipv6 MY_IPV6_ACL in
Removing application of an ACL from a VLAN	<u>apply access-list</u>	vlan 40 no apply access-list mac MY_MAC_ACL in
Removing application of an ACL from the Control Plane (OOBM)	<u>apply access-list control-plane</u>	no apply access-list ip MY_IP_ACL control-plane vrf mgmt
Showing all ACLs	<u>show access-list</u>	show access-list
Showing all IPv6 ACLs	<u>show access-list</u>	show access-list ipv6
Showing all ACLs applied to interface 1/1/1	<u>show access-list</u>	show access-list interface 1/1/1
Showing all ACLs applied to VLAN 10	<u>show access-list</u>	show access-list vlan 10

Table Continued

Task	Command or procedure	Example
Showing all ACLs applied to the Control Plane	<u>show access-list control-plane</u>	show access-list control-plane
Showing a particular ACL	<u>show access-list</u>	show access-list ip MY_ACL
Showing an ACL as commands	<u>show access-list</u>	show access-list ip MY_ACL commands
Showing ACL hit counts for an ACL applied to an interface	<u>show access-list hitcounts</u>	show access-list hitcounts ip MY_ACL interface 1/1/1
Showing ACL hit counts for an ACL applied to a VLAN	<u>show access-list hitcounts</u>	show access-list hitcounts ip MY_ACL vlan 10
Showing ACL hit counts for an ACL applied to the Control Plane	<u>show access-list hitcounts control-plane</u>	show access-list hitcounts ip MY_ACL control-plane vrf mgmt
Clearing ACL hit counts	<u>clear access-list hitcounts</u>	clear access-list hitcounts ip MY_ACL vlan 10
Clearing ACL hit counts for Control Plane	<u>clear access-list hitcounts control-plane</u>	clear access-list hitcounts control-plane vrf mgmt
Copying an ACL	<u>access-list copy</u>	access-list ipv6 MY_IPV6_ACL copy MY_IPV6_ACL2
Resequencing the ACEs of an ACL	<u>access-list resequence</u>	access-list ip MY_IP_ACL resequence 1 1
Resetting an ACL	<u>access-list reset</u>	access-list ip MY_IP_ACL reset
Setting the ACL log timer frequency	<u>access-list log-timer</u>	access-list log-timer 30

Object group-related tasks

Object groups are useful for defining groups of addresses and ports for use exclusively in the two ACL-defining commands `access-list ip` and `access-list ipv6`.

Common object group-related tasks are as follows. Simple tasks link to the relevant command description.

Task	Command or procedure	Example
Creating an IPv4 address object group	<u>object-group ip address</u>	object-group ip address my_ipv4_addr_group
Creating an IPv6 address object group	<u>object-group ipv6 address</u>	object-group ipv6 address my_ipv6_addr_group

Table Continued

Task	Command or procedure	Example
Creating a port object group	<u>object-group port</u>	object-group port my_port_group
Showing an IPv4 address object group	<u>show object-group</u>	show object-group ip address my_ipv4_addr_group
Showing all IPv6 address object groups	<u>show object-group</u>	show object-group ipv6 address
Showing a port object group	<u>show object-group</u>	show object-group port my_port_group
Showing all port object groups as commands	<u>show object-group</u>	show object-group port commands
Resequencing an IPv4 address object group	<u>object-group ip address</u>	object-group ip address my_ipv4_addr_group resequence 100 10
Resequencing a port object group	<u>object-group port</u>	object-group port my_port_group resequence 200 5
Resetting an IPv6 address object group	<u>object-group ipv6 address</u>	object-group ipv6 address my_ipv6_addr_group reset
Resetting a port object group	<u>object-group port</u>	object-group port my_port_group reset

ACL considerations

Active ACL configuration versus user-specified configuration

The `show access-list` command shows the active configuration of the switch. The active configuration is the ACLs that have been configured and accepted by the system. The active configurations are the interfaces on which the ACLs have successfully been programmed in the hardware.

The output of the `show access-list` command with the `configuration` parameter shows the ACLs that have been configured. The output of this command may not be the same as what was programmed in the hardware or what is active on the switch. The situation might occur because of one or more of the following:

- Unsupported command parameters might have been configured.
- Unsupported applications might have been specified.
- Applying an ACL might have been unsuccessful due to lack of hardware resources.

To determine if a discrepancy exists between what was configured and what is active, run the `show access-list` command with the `configuration` parameter.

If the active ACLs and configured ACLs are not the same, the switch shows a warning message in the output of the `show` command:

```
! access-list ip MY_IP ACL user configuration does not match active configuration.
! run 'access-list TYPE NAME reset' to reset access-list to match active configuration.
```

If the configured ACL is processing, the switch shows an in-progress warning.

```
! access-list ip MY_IP_ACL user configuration currently being processed
! run 'access-list TYPE NAME reset' to reset access-list to match active configuration.
```

If the switch shows a warning message or in-progress message, additional changes can be made until the error message is no longer shown in the show command, or you can run the `access-list {all|ip <ACL-NAME>|ipv6 <ACL-NAME>|mac <ACL-NAME>} reset` command. The `access-list reset` command changes the user-specified configuration to match the active configuration. For details, see [access-list reset](#).



NOTE: The `show running-config` command also shows a warning about ACLs that are in progress or failed.

Examples

Applying an ACL with TCP acknowledgments (ACKs) on egress:

```
switch(config-acl)# 10 permit tcp 172.16.2.0/16 any ack
```

Showing the user-specified configuration:

```
switch(config)# do show access-list ip TEST_ACL
    10 permit tcp 172.16.2.0/16 any ack
interface 1/1/1
! access-list ip TEST_ACL user configuration does not match active configuration.
! run 'show access-list [commands]' to display active access-list configuration.
apply access-list ip TEST_ACL out

switch(config)# do show access-list commands
access-list ip TEST_ACL
    10 permit tcp 172.16.2.0/16 any ack
! access-list ip TEST_ACL user configuration does not match active configuration.
! run 'access-list all reset' to reset all access-lists to match active configuration.

switch(config)# do show access-list commands configuration
access-list ip TEST_ACL
    10 permit tcp 172.16.2.0/16 any ack
! access-list ip TEST_ACL user configuration does not match active configuration.
! run 'access-list all reset' to reset all access-lists to match active configuration.
interface 1/1/1
apply access-list ip TEST_ACL out

switch(config)# do show access-list
Type      Name
Sequence  Comment
          Action
          Source IP Address
          Destination IP Address
          Additional Parameters
-----
IPv4      TEST_ACL
          10 permit
          172.16.2.0/16
          any
          ack
                                     L3 Protocol
                                     Source L4 Port(s)
                                     Destination L4 Port(s)
```

Resetting the user-specified configuration to match the active configuration:

```
switch(config)# access-list all reset
```

Showing the updated user-specified configuration:

```
switch(config)# do show access-list commands configuration
access-list ip TEST ACL
10 permit tcp 172.16.2.0/16 any ack
```

ACL commands

ACL application

ACLs can be applied as follows:

ACL type	IPv4+6	IPv4+6	MAC	MAC
Direction	In	Out	In	Out
L2 interface (port)	Yes	Yes	Yes	Yes
L2 LAG	Yes	Yes	Yes	Yes
VLAN	Yes	Yes	Yes	Yes
Interface VLAN	Yes (routed)	Yes (routed)		
Management interface	Yes			
Control plane (per VRF)	Yes			



NOTE: The following match criteria is not supported. If this match criteria is attempted to be configured, an error message will be displayed and the action will not be completed.

TTL on IP ACLs



NOTE: To apply IPv4 and/or IPv6 ACLs to the management interface, apply them to the control plane on the management VRF.

access-list copy

Syntax

```
access-list {ip|ipv6|mac} <ACL-NAME> copy <DESTINATION-ACL>
```

Description

Copies an IPv4, IPv6, or MAC ACL to a new destination ACL or overwrites an existing ACL.

Command context

config

Parameters

{ip|ipv6|mac}

Specifies the type of ACL.

<ACL-NAME>

Specifies the name of the ACL to be copied.

<DESTINATION-ACL>

Specifies the name of the destination ACL.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Copying MY_IP_ACL to MY_IP_ACL2:

```
switch(config)# access-list ip MY_IP_ACL copy MY_IP_ACL2
switch(config-acl-ip)# exit
```

```
switch(config)# do show access-list
```

Type	Name		
Sequence	Comment	Action	L3 Protocol
		Source IP Address	Source L4 Port(s)
		Destination IP Address	Destination L4 Port(s)
		Additional Parameters	

IPv4	MY_IP_ACL		
1	permit		udp
	any	172.16.1.0/255.255.255.0	
2	permit		tcp
	172.16.2.0/255.255.0.0		> 1023
	any		
3	permit		tcp
	172.26.1.0/255.255.255.0		
	any		
	dscp: AF11		
	ack		
	syn		
4	deny		any
	any		
	any		
	Hit-counts: enabled		

IPv4	MY_IP_ACL2		
1	permit		udp
	any	172.16.1.0/255.255.255.0	
2	permit		tcp
	172.16.2.0/255.255.0.0		> 1023
	any		
3	permit		tcp
	172.26.1.0/255.255.255.0		
	any		
	dscp: AF11		
	ack		
	syn		
4	deny		any
	any		
	any		
	Hit-counts: enabled		

Copying MY_IPV6_ACL to MY_IPV6_ACL2:

```
switch(config)# access-list ipv6 MY_IPV6_ACL copy MY_IPV6_ACL2
switch(config-acl-ip)# exit
```

```
switch(config)# do show access-list
```

Type	Name		
Sequence	Comment	Action	L3 Protocol

	Source IP Address	Source L4 Port(s)
	Destination IP Address	Destination L4 Port(s)
	Additional Parameters	

IPv6	MY_IPV6_ACL	
1	permit any 2001::1/64	udp
2	Permit all TCP ephemeral ports permit 2001:2001::2:1 any	tcp > 1023
3	permit 2001:2011::1/64 any	tcp
4	deny any any Hit-counts: enabled	any

IPv6	MY_IPV6_ACL2	
1	permit any 2001::1/64	udp
2	Permit all TCP ephemeral ports permit 2001:2001::2:1 any	tcp > 1023
3	permit 2001:2011::1/64 any	tcp
4	deny any any Hit-counts: enabled	any

Copying MY_MAC_ACL to MY_MAC_ACL2:

```
switch(config)# access-list mac MY_MAC_ACL copy MY_MAC_ACL2
switch(config-acl-mac)# exit
```

```
switch(config)# do show access-list
```

Type	Name	
Sequence	Comment	
	Action	EtherType
	Source MAC Address	
	Destination MAC Address	
	Additional Parameters	

MAC	MY_MAC_ACL	
1	permit 1122.3344.5566/ffff.ffff.0000 any	ipv6
2	permit aaaa.bbbb.cccc 1111.2222.3333 QoS Priority Code Point: 4	any
3	Permit all vlan-1 tagged Appletalk traffic permit any any VLAN: 1	appletalk
4	deny any	any

```

any
Hit-counts: enabled
-----
MAC      MY_MAC_ACL2
1 permit 1122.3344.5566/ffff.ffff.0000      ipv6
any
2 permit aaaa.bbbb.cccc                    any
1111.2222.3333
QoS Priority Code Point: 4
3 Permit all vlan-1 tagged Appletalk traffic
permit appletalk
any
any
VLAN: 1
4 deny any
any
Hit-counts: enabled

```

access-list ip

Syntax

Syntax to create an IPv4 ACL and enter its context. Plus syntax to remove an ACL:

```
access-list ip <ACL-NAME>
```

```
no access-list ip <ACL-NAME>
```

Syntax (within the ACL context) for creating or removing ACEs for protocols ah, gre, esp, igmp, ospf, pim (ip is available as an alias for any):

```

[ <SEQUENCE-NUMBER> ]
{permit|deny}
{any|ip|ah|gre|esp|igmp|ospf|pim|<IP-PROTOCOL-NUM>}
{any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}] |<ADDRESS-GROUP>}
{any|<DST-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}] |<ADDRESS-GROUP>}
[dscp <DSCP-SPECIFIER>] [ecn <ECN-VALUE>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [ttl <TTL-VALUE>] [count] [log]

```

```
no <SEQUENCE-NUMBER>
```

Syntax (within the ACL context) for creating or removing ACEs for protocols sctp, tcp, udp:

```

[ <SEQUENCE-NUMBER> ]
{permit|deny}
{sctp|tcp|udp}
{any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}] |<ADDRESS-GROUP>}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>|group <PORT-GROUP>]
{any|<DST-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}] |<ADDRESS-GROUP>}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>|group <PORT-GROUP>]
[urg] [ack] [psh] [rst] [syn] [fin] [established]
[dscp <DSCP-SPECIFIER>] [ecn <ECN-VALUE>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [ttl <TTL-VALUE>] [count] [log]

```

```
no <SEQUENCE-NUMBER>
```

Syntax (within the ACL context) for creating or removing ACEs for protocol icmp:

```
[<SEQUENCE-NUMBER>]
{permit|deny}
{icmp}
{any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>} ] |<ADDRESS-GROUP>}
{any|<DST-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>} ] |<ADDRESS-GROUP>}
[icmp-type {echo|echo-reply|<ICMP-TYPE-VALUE>}] [icmp-code <ICMP-CODE-VALUE>]
[dscp <DSCP-SPECIFIER>] [ecn <ECN-VALUE>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [ttl <TTL-VALUE>] [count] [log]

no <SEQUENCE-NUMBER>
```

Syntax (within the ACL context) for ACE comments:

```
[<SEQUENCE-NUMBER>] comment <TEXT-STRING>

no <SEQUENCE-NUMBER> comment
```

Description

Creates an IPv4 Access Control List (ACL) comprised of one or more Access Control Entries (ACEs) ordered and prioritized by sequence number. The lowest sequence number is the highest prioritized ACE.

The `no` form of this command deletes the entire ACL, or deletes an ACE identified by sequence number, or deletes only the comment from the ACE identified by sequence number.

Command context

config

The `access-list ip <ACL-NAME>` command takes you into the named ACL context where you enter the ACEs.

Parameters

<ACL-NAME>

Specifies the name of this ACL.

<SEQUENCE-NUMBER>

Specifies a sequence number for the ACE. Range: 1 to 4294967295.

{permit|deny}

Specifies whether to permit or deny traffic matching this ACE.

<IP-PROTOCOL-NUM>

Specifies the protocol as its Internet Protocol number. For example, 2 corresponds to the IGMP protocol. Range: 0 to 255.

{any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}] |<ADDRESS-GROUP>}

Specifies the source IPv4 address.

- `any` - specifies any source IPv4 address.
- `<SRC-IP-ADDRESS>` - specifies the source IPv4 host address.

- `<PREFIX-LENGTH>` - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32.
- `<SUBNET-MASK>` - specifies the address bits to mask (dotted decimal notation).
- `<ADDRESS-GROUP>` - specifies an IPv4 address group defined with `object-group ip address`.

{ any | `<DST-IP-ADDRESS>` [/ { `<PREFIX-LENGTH>` | `<SUBNET-MASK>` }] | `<ADDRESS-GROUP>` }

Specifies the destination IPv4 address.

- `any` - specifies any destination IPv4 address.
- `<DST-IP-ADDRESS>` - specifies the destination IPv4 host address.
 - `<PREFIX-LENGTH>` - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32.
 - `<SUBNET-MASK>` - specifies the address bits to mask (dotted decimal notation).
- `<ADDRESS-GROUP>` - specifies an IPv4 address group that you defined earlier with `object-group ip address`.

[{ `eq` | `gt` | `lt` } `<PORT>` | `range` `<MIN-PORT>` `<MAX-PORT>` | `group` `<PORT-GROUP>`]

Specifies the port, port range, or port group. Port numbers are in the range of 0 to 65535.

- `eq` `<PORT>` - specifies the Layer 4 port.
- `gt` `<PORT>` - specifies any Layer 4 port greater than the indicated port.
- `lt` `<PORT>` - specifies any Layer 4 port less than the indicated port.
- `range` `<MIN-PORT>` `<MAX-PORT>` - specifies the Layer 4 port range.
- `group` `<PORT-GROUP>` - specifies the Layer 4 port group that you defined earlier with `object-group port`.



NOTE: Upon application of the ACL, ACEs with L4 port ranges may consume more than one hardware entry.

urg

Specifies matching on the TCP Flag: Urgent.

ack

Specifies matching on the TCP Flag: Acknowledgment.

psh

Specifies matching on the TCP Flag: Push buffered data to receiving application.

rst

Specifies matching on the TCP Flag: Reset the connection.

syn

Specifies matching on the TCP Flag: Synchronize sequence numbers.

fin

Specifies matching on the TCP Flag: Finish connection.

established

Specifies matching on the TCP Flag: `Established` connection.

[icmp-type {echo|echo-reply|<ICMP-TYPE-VALUE>}]

Specifies the ICMP type.

- `echo` - specifies an ICMP echo request packet.
- `echo-reply` - specifies an ICMP echo reply packet.
- `<ICMP-TYPE-VALUE>` - specifies an ICMP type value. Range: 0 to 255.

[icmp-code <ICMP-CODE-VALUE>]

Specifies the ICMP code value. Range: 0 to 255.

dscp <DSCP-SPECIFIER>

Specifies the Differentiated Services Code Point (DSCP), either a numeric `<DSCP-VALUE>` (0 to 63) or one of these keywords:

- `AF11` - DSCP 10 (Assured Forwarding Class 1, low drop probability)
- `AF12` - DSCP 12 (Assured Forwarding Class 1, medium drop probability)
- `AF13` - DSCP 14 (Assured Forwarding Class 1, high drop probability)
- `AF21` - DSCP 18 (Assured Forwarding Class 2, low drop probability)
- `AF22` - DSCP 20 (Assured Forwarding Class 2, medium drop probability)
- `AF23` - DSCP 22 (Assured Forwarding Class 2, high drop probability)
- `AF31` - DSCP 26 (Assured Forwarding Class 3, low drop probability)
- `AF32` - DSCP 28 (Assured Forwarding Class 3, medium drop probability)
- `AF33` - DSCP 30 (Assured Forwarding Class 3, high drop probability)
- `AF41` - DSCP 34 (Assured Forwarding Class 4, low drop probability)
- `AF42` - DSCP 36 (Assured Forwarding Class 4, medium drop probability)
- `AF43` - DSCP 38 (Assured Forwarding Class 4, high drop probability)
- `CS0` - DSCP 0 (Class Selector 0: Default)
- `CS1` - DSCP 8 (Class Selector 1: Scavenger)
- `CS2` - DSCP 16 (Class Selector 2: OAM)
- `CS3` - DSCP 24 (Class Selector 3: Signaling)
- `CS4` - DSCP 32 (Class Selector 4: Real time)
- `CS5` - DSCP 40 (Class Selector 5: Broadcast video)
- `CS6` - DSCP 48 (Class Selector 6: Network control)
- `CS7` - DSCP 56 (Class Selector 7)
- `EF` - DSCP 46 (Expedited Forwarding)

ecn *<ECN-VALUE>*

Specifies an Explicit Congestion Notification value. Range: 0 to 3.

ip-precedence *<IP-PRECEDENCE-VALUE>*

Specifies an IP precedence value. Range: 0 to 7.

tos *<TOS-VALUE>*

Specifies the Type of Service value. Range: 0 to 31.

fragment

Specifies a fragment packet.

vlan *<VLAN-ID>*

Specifies VLAN tag to match on. 802.1Q VLAN ID.



NOTE: This parameter cannot be used in any ACL that will be applied to a VLAN.

ttl *<TTL-VALUE>*

Specifies a time-to-live (hop limit) value. Range: 0 to 255. Not supported for ACLs.

count

Keeps the hit counts of the number of packets matching this ACE.

log

Keeps a log of the number of packets matching this ACE. Works with both `permit` and `deny` actions. Works with ACLs applied on ingress or egress, except for control plane.

[*<SEQUENCE-NUMBER>*] **comment** *<TEXT-STRING>*

Adds a comment to an ACE. The `no` form removes only the comment from the ACE.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- If the *<IP-PROTOCOL-NUM>* parameter is used instead of a protocol name, ensure that any needed ACE-definition parameters specific to the selected protocol are also provided.
- When using multiple ACL types (IPv4, IPv6, or MAC) with logging on the same interface, the first packet that matches an ACE with `log` option is logged. Until the log-timer wait-period is over, any packets matching other ACL types do not create a log. At the end of the wait-period, the switch creates a summary log for all the ACLs that were matched, regardless of type.

Examples

Creating an IPv4 ACL with four entries:

```
switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# 10 permit udp any 172.16.1.0/24
switch(config-acl-ip)# 20 permit tcp 172.16.2.0/16 gt 1023 any
switch(config-acl-ip)# 30 permit tcp 172.26.1.0/24 any syn ack dscp 10
switch(config-acl-ip)# 40 deny any any any count
switch(config-acl-ip)# exit

switch(config)# show access-list
Type          Name
```

Sequence	Comment	Action	Source IP Address	Destination IP Address	Additional Parameters	L3 Protocol	Source L4 Port(s)	Destination L4 Port(s)
IPv4	MY_IP_ACL							
10		permit	any	172.16.1.0/255.255.255.0		udp		
20		permit	any	172.16.2.0/255.255.0.0		tcp	> 1023	
30		permit	any	172.26.1.0/255.255.255.0		tcp		
				dscp: AF11				
				ack				
				syn				
40		deny	any	any		any		
				Hit-counts: enabled				

Adding a comment to an existing IPv4 ACE:

```
switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# 20 comment Permit all TCP ephemeral ports
switch(config-acl-ip)# exit
```

```
switch(config)# show access-list
```

Type	Sequence	Comment	Action	Source IP Address	Destination IP Address	Additional Parameters	L3 Protocol	Source L4 Port(s)	Destination L4 Port(s)
IPv4		MY_IP_ACL							
	10		permit	any	172.16.1.0/255.255.255.0		udp		
	20	Permit all TCP ephemeral ports	permit	any	172.16.2.0/255.255.0.0		tcp	> 1023	
	30		permit	any	172.26.1.0/255.255.255.0		tcp		
				dscp: AF11					
				ack					
				syn					
	40		deny	any	any		any		
				Hit-counts: enabled					

Removing a comment from an existing IPv4 ACE:

```
switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# no 20 comment
switch(config-acl-ip)# exit
```

```
switch(config)# show access-list
```

Type	Name		
Sequence	Comment	Action	L3 Protocol
		Source IP Address	Source L4 Port(s)
		Destination IP Address	Destination L4 Port(s)
		Additional Parameters	

IPv4	MY_IP_ACL		
10	permit		udp
	any		
	172.16.1.0/255.255.255.0		
20	permit		tcp
	172.16.2.0/255.255.0.0		> 1023
	any		
30	permit		tcp
	172.26.1.0/255.255.255.0		
	any		
	dscp: AF11		
	ack		
	syn		
40	deny		any
	any		
	any		
	Hit-counts: enabled		

Adding an ACE (insert line 25) to an existing IPv4 ACL:

```
switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# 25 permit icmp 172.16.2.0/16 any
switch(config-acl-ip)# exit
```

```
switch(config)# show access-list
```

Type	Name		
Sequence	Comment	Action	L3 Protocol
		Source IP Address	Source L4 Port(s)
		Destination IP Address	Destination L4 Port(s)
		Additional Parameters	

IPv4	MY_IP_ACL		
10	permit		udp
	any		
	172.16.1.0/255.255.255.0		
20	permit		tcp
	172.16.2.0/255.255.0.0		> 1023
	any		
25	permit		icmp
	172.16.2.0/255.255.0.0 any		
30	permit		tcp
	172.26.1.0/255.255.255.0		
	any		
	dscp: AF11		
	ack		
	syn		
40	deny		any
	any		
	any		
	Hit-counts: enabled		

Replacing an ACE in an existing IPv4 ACL:

```
switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# 25 permit icmp 172.17.1.0/16 any
switch(config-acl-ip)# exit
```



```

switch(config)# show access-list
Type          Name
Sequence      Comment
              Action
              Source IP Address
              Destination IP Address
              Additional Parameters
-----
IPv4          MY_IP_ACL
10 permit
   any
   172.16.1.0/255.255.255.0
20 permit
   172.16.2.0/255.255.0.0
   any
25 permit
   172.17.1.0/255.255.0.0
30 permit
   172.26.1.0/255.255.255.0
   any
   dscp: AF11
   ack
   syn
40 deny
   any
   any
Hit-counts: enabled
L3 Protocol
Source L4 Port(s)
Destination L4 Port(s)
udp
tcp
> 1023
icmp
tcp
any

```

Removing an ACE from an IPv4 ACL:

```

switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# no 25
switch(config-acl-ip)# exit

switch(config)# show access-list
Type          Name
Sequence      Comment
              Action
              Source IP Address
              Destination IP Address
              Additional Parameters
-----
IPv4          MY_IP_ACL
10 permit
   any
   172.16.1.0/255.255.255.0
20 permit
   172.16.2.0/255.255.0.0
   any
30 permit
   172.26.1.0/255.255.255.0
   any
   dscp: AF11
   ack
   syn
40 deny
   any
   any
Hit-counts: enabled
L3 Protocol
Source L4 Port(s)
Destination L4 Port(s)
udp
tcp
> 1023
tcp
any

```

Copy an IPv4 ACL:

```

switch(config)# access-list ip MY_IP_ACL copy MY_IP_ACL2
switch(config)# show access-list
Type      Name
Sequence Comment
Action
Source IP Address
Destination IP Address
Additional Parameters
L3 Protocol
Source L4 Port(s)
Destination L4 Port(s)
-----
IPv4      MY_IP_ACL
10
  permit
  any
  172.16.1.0/255.255.255.0
  udp
20
  permit
  172.16.2.0/255.255.0.0
  > 1023
  tcp
30
  permit
  172.26.1.0/255.255.255.0
  tcp
  any
  dscp: AF11
  ack
  syn
40
  deny
  any
  any
  Hit-counts: enabled
-----
IPv4      MY_IP_ACL2
10
  permit
  any
  172.16.1.0/255.255.255.0
  udp
20
  permit
  172.16.2.0/255.255.0.0
  > 1023
  tcp
30
  permit
  172.26.1.0/255.255.255.0
  tcp
  any
  dscp: AF11
  ack
  syn
40
  deny
  any
  any
  Hit-counts: enabled

```

Removing an IPv4 ACL:

```

switch(config)# no access-list ip MY_IP_ACL

switch(config)# show access-list
Type      Name
Sequence Comment
Action
Source IP Address
Destination IP Address
Additional Parameters
L3 Protocol
Source L4 Port(s)
Destination L4 Port(s)

```

```

IPv4      MY_IP_ACL2
1 permit                                udp
   any
   172.16.1.0/255.255.255.0
2 permit                                tcp
   172.16.2.0/255.255.0.0
   > 1023
   any
3 permit                                tcp
   172.26.1.0/255.255.255.0
   any
   dscp: AF11
   ack
   syn
4 deny                                    any
   any
   any
Hit-counts: enabled

```

access-list ipv6

Syntax

Syntax to create an IPv6 ACL and enter its context. Plus syntax to remove an ACL:

```
access-list ipv6 <ACL-NAME>
```

```
no access-list ipv6 <ACL-NAME>
```

Syntax (within the ACL context) for creating or removing ACEs for protocols ah, gre, esp, ospf, pim (ipv6 is available as an alias for any):

```

[<SEQUENCE-NUMBER>]
{permit|deny}
{any|ipv6|ah|gre|esp|ospf|pim|<IP-PROTOCOL-NUM>}
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>]|<ADDRESS-GROUP>}
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>]|<ADDRESS-GROUP>}
[dscp <DSCP-SPECIFIER>] [ecn <ECN-VALUE>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [ttl <TTL-VALUE>] [count] [log]

```

```
no <SEQUENCE-NUMBER>
```

Syntax (within the ACL context) for creating or removing ACEs for protocols sctp, tcp, udp:

```

[<SEQUENCE-NUMBER>]
{permit|deny}
{sctp|tcp|udp}
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>]|<ADDRESS-GROUP>}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>|group <PORT-GROUP>]
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>]|<ADDRESS-GROUP>}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>|group <PORT-GROUP>]
[urg] [ack] [psh] [rst] [syn] [fin] [established]
[dscp <DSCP-SPECIFIER>] [ecn <ECN-VALUE>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [ttl <TTL-VALUE>] [count] [log]

```

```
no <SEQUENCE-NUMBER>
```

Syntax (within the ACL context) for creating or removing ACEs for protocol icmpv6:

```

[<SEQUENCE-NUMBER>]
{permit|deny}
{icmpv6}

```

```
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>]|<ADDRESS-GROUP>}
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>]|<ADDRESS-GROUP>}
[icmp-type {echo|echo-reply|<ICMP-TYPE-VALUE>}] [icmp-code <ICMP-CODE-VALUE>]
[dscp <DSCP-SPECIFIER>] [ecn <ECN-VALUE>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [ttl <TTL-VALUE>] [count] [log]

no <SEQUENCE-NUMBER>
```

Syntax (within the ACL context) for ACE comments:

```
[<SEQUENCE-NUMBER>] comment <TEXT-STRING>

no <SEQUENCE-NUMBER> comment
```

Description

Creates an IPv6 Access Control List (ACL). The ACL is made of one or more Access Control Entries (ACEs) ordered and prioritized by sequence number. The lowest sequence number is the highest prioritized ACE.

The **no** form of this command deletes the entire ACL, or deletes an ACE identified by sequence number, or deletes only the comment from the ACE identified by sequence number.

Command context

```
config
```

The **access-list ipv6 <ACL-NAME>** command takes you into the named ACL context where you enter the ACEs.

Parameters

<ACL-NAME>

Specifies the name of this ACL.

<SEQUENCE-NUMBER>

Specifies a sequence number for the ACE. Range: 1 to 4294967295.

{permit|deny}

Specifies whether to permit or deny traffic matching this ACE.

<IP-PROTOCOL-NUM>

Specifies the protocol as its Internet Protocol number. For example, 2 corresponds to the IGMP protocol. Range: 0 to 255.

{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>]|<ADDRESS-GROUP>}

Specifies the source IPv6 address.

- **any** - specifies any source IPv6 address.
- **<SRC-IP-ADDRESS>** - specifies the source IPv6 host address.
 - **<PREFIX-LENGTH>** - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 128.
- **<ADDRESS-GROUP>** - specifies an IPv6 address group that you defined earlier with **object-group ipv6** address.

{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>]|<ADDRESS-GROUP>}

Specifies the destination IPv6 address.

- any - specifies any destination IPv6 address.
- `<DST-IP-ADDRESS>` - specifies the destination IPv6 host address.
 - `<PREFIX-LENGTH>` - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 128.
- `<ADDRESS-GROUP>` - specifies an IPv6 address group that you defined earlier with `object-group ipv6 address`.

[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>|group <PORT-GROUP>]

Specifies the port, port range, or port group. Port numbers are in the range of 0 to 65535.

- eq <PORT> - specifies the Layer 4 port.
- gt <PORT> - specifies any Layer 4 port greater than the indicated port.
- lt <PORT> - specifies any Layer 4 port less than the indicated port.
- range <MIN-PORT> <MAX-PORT> - specifies the Layer 4 port range.
- group <PORT-GROUP> - specifies the Layer 4 port group that you defined earlier with `object-group port`.



NOTE: Upon application of the ACL, ACEs with L4 port ranges may consume more than one hardware entry.

urg, ack, psh, rst, syn, fin, established

These TCP flag-matching parameters are supported for both ingress and egress.

[icmp-type {echo|echo-reply|<ICMP-TYPE-VALUE>}]

Specifies the ICMP type.

- echo - specifies an ICMP echo request packet.
- echo-reply - specifies an ICMP echo reply packet.
- `<ICMP-TYPE-VALUE>` - specifies an ICMP type value. Range: 0 to 255.

[icmp-code <ICMP-CODE-VALUE>]

Specifies the ICMP code value. Range: 0 to 255.

dscp <DSCP-SPECIFIER>

Specifies the Differentiated Services Code Point (DSCP), either a numeric `<DSCP-VALUE>` (0 to 63) or one of these keywords:

- AF11 - DSCP 10 (Assured Forwarding Class 1, low drop probability)
- AF12 - DSCP 12 (Assured Forwarding Class 1, medium drop probability)
- AF13 - DSCP 14 (Assured Forwarding Class 1, high drop probability)
- AF21 - DSCP 18 (Assured Forwarding Class 2, low drop probability)
- AF22 - DSCP 20 (Assured Forwarding Class 2, medium drop probability)
- AF23 - DSCP 22 (Assured Forwarding Class 2, high drop probability)

- AF31 - DSCP 26 (Assured Forwarding Class 3, low drop probability)
- AF32 - DSCP 28 (Assured Forwarding Class 3, medium drop probability)
- AF33 - DSCP 30 (Assured Forwarding Class 3, high drop probability)
- AF41 - DSCP 34 (Assured Forwarding Class 4, low drop probability)
- AF42 - DSCP 36 (Assured Forwarding Class 4, medium drop probability)
- AF43 - DSCP 38 (Assured Forwarding Class 4, high drop probability)
- CS0 - DSCP 0 (Class Selector 0: Default)
- CS1 - DSCP 8 (Class Selector 1: Scavenger)
- CS2 - DSCP 16 (Class Selector 2: OAM)
- CS3 - DSCP 24 (Class Selector 3: Signaling)
- CS4 - DSCP 32 (Class Selector 4: Real time)
- CS5 - DSCP 40 (Class Selector 5: Broadcast video)
- CS6 - DSCP 48 (Class Selector 6: Network control)
- CS7 - DSCP 56 (Class Selector 7)
- EF - DSCP 46 (Expedited Forwarding)

ecn *<ECN-VALUE>*

Specifies an Explicit Congestion Notification value. Range: 0- 3.

ip-precedence *<IP-PRECEDENCE-VALUE>*

Specifies an IP precedence value. Range: 0-7.

tos *<TOS-VALUE>*

Specifies the Type of Service value. Range: 0-31.

fragment

Specifies a fragment packet.

vlan *<VLAN-ID>*

Specifies VLAN tag to match on. 802.1Q VLAN ID.



NOTE: This parameter cannot be used in any ACL that will be applied to a VLAN.

ttl *<TTL-VALUE>*

Not supported.

count

Keeps the hit counts of the number of packets matching this ACE.

log

Keeps a log of the number of packets matching this ACE. Works with both `permit` and `deny` actions. Works with ACLs applied on ingress or egress, except for control plane.

[<SEQUENCE-NUMBER>] comment *<TEXT-STRING>*

Adds a comment to an ACE. The `no` form removes only the comment from the ACE.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- If the `<IP-PROTOCOL-NUM>` parameter is used instead of a protocol name, ensure that any needed ACE-definition parameters specific to the selected protocol are also provided.
- When using multiple ACL types (IPv4, IPv6, or MAC) with logging on the same interface, the first packet that matches an ACE with `log` option is logged. Until the log-timer wait-period is over, any packets matching other ACL types do not create a log. At the end of the wait-period, the switch creates a summary log all the ACLs that were matched, regardless of type.

Examples

Creating an IPv6 ACL with four entries:

```
switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# 10 permit udp any 2001::1/64
switch(config-acl-ipv6)# 20 permit tcp 2001:2001::2:1/128 gt 1023 any
switch(config-acl-ipv6)# 30 permit tcp 2001:2011::1/64 any
switch(config-acl-ipv6)# 40 deny any any any count
switch(config-acl-ipv6)# exit
```

```
switch(config)# do show access-list
```

Type	Name			
Sequence	Comment	Action	L3 Protocol	
		Source IP Address	Source L4 Port(s)	
		Destination IP Address	Destination L4 Port(s)	
		Additional Parameters		

IPv6	MY_IPV6_ACL			
10	permit		udp	
	any			
	2001::1/64			
20	permit		tcp	
	2001:2001::2:1		> 1023	
	any			
30	permit		tcp	
	2001:2011::1/64			
	any			
40	deny		any	
	any			
	any			
	Hit-counts: enabled			

Adding a comment to an existing IPv6 ACE:

```
switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# 20 comment Permit all TCP ephemeral ports
switch(config-acl-ipv6)# exit
```

```
switch(config)# do show access-list
```

Type	Name			
Sequence	Comment	Action	L3 Protocol	
		Source IP Address	Source L4 Port(s)	
		Destination IP Address	Destination L4 Port(s)	
		Additional Parameters		

IPv6	MY_IPV6_ACL			

```

10 permit                                udp
   any
   2001::1/64
20 Permit all TCP ephemeral ports
   permit                                tcp
   2001:2001::2:1                        > 1023
   any
30 permit                                tcp
   2001:2011::1/64
   any
40 deny                                  any
   any
   any
Hit-counts: enabled

```

Removing a comment from an existing IPv6 ACE:

```

switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# no 20 comment
switch(config-acl-ipv6)# exit

```

```

switch(config)# do show access-list

```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address	Destination L4 Port(s)	Additional Parameters
IPv6	MY_IPV6_ACL	10		permit	udp	any		2001::1/64		
		20		permit	tcp	2001:2001::2:1	> 1023	any		
		30		permit	tcp	2001:2011::1/64		any		
		40		deny	any	any		any		
				any						
				any						
				Hit-counts: enabled						

Adding an ACE to an existing IPv6 ACL:

```

switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# 25 permit icmpv6 2001::1/64 any
switch(config-acl-ipv6)# exit

```

```

switch(config)# do show access-list

```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address	Destination L4 Port(s)	Additional Parameters
IPv6	MY_IPV6_ACL	10		permit	udp	any		2001::1/64		
		20		permit	tcp	2001:2001::2:1	> 1023			


```

    any
25 permit                                icmpv6
   2001::1/64
    any
30 permit                                tcp
   2001:2011::1/64
    any
40 deny                                  any
   any
   any
Hit-counts: enabled

```

Replacing an ACE in an existing IPv6 ACL:

```

switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# 25 permit icmpv6 2001::2:1/64 any
switch(config-acl-ipv6)# exit

```

```

switch(config)# do show access-list

```

Type	Name			
Sequence	Comment	Action	L3 Protocol	
		Source IP Address	Source L4 Port(s)	
		Destination IP Address	Destination L4 Port(s)	
		Additional Parameters		
IPv6	MY_IPV6_ACL			
10	permit		udp	
	any			
	2001::1/64			
20	permit		tcp	
	2001:2001::2:1		> 1023	
	any			
25	permit		icmpv6	
	2001::2:1/64			
	any			
30	permit		tcp	
	2001:2011::1/64			
	any			
40	deny		any	
	any			
	any			
	Hit-counts: enabled			

Removing an ACE from an IPv6 ACL:

```

switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# no 25
switch(config-acl-ipv6)# exit

```

```

switch(config)# do show access-list

```

Type	Name			
Sequence	Comment	Action	L3 Protocol	
		Source IP Address	Source L4 Port(s)	
		Destination IP Address	Destination L4 Port(s)	
		Additional Parameters		
IPv6	MY_IPV6_ACL			
10	permit		udp	
	any			
	2001::1/64			
20	permit		tcp	
	2001:2001::2:1		> 1023	

```

    any
30 permit                                tcp
    2001:2011::1/64
    any
40 deny                                any
    any
    any
Hit-counts: enabled

```

Removing an IPv6 ACL:

```
switch(config)# no access-list ipv6 MY_IPV6_ACL
```

```
switch(config)# do show access-list
```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address	Destination L4 Port(s)	Additional Parameters
IPv6	MY_IPV6_ACL2	1	permit		udp	any		2001::1/64		
		2	Permit all TCP ephemeral ports	permit	tcp	2001:2001::2:1	> 1023	any		
		3	permit		tcp	2001:2011::1/64		any		
		4	deny		any	any		any		
										Hit-counts: enabled

access-list log-timer

Syntax

```
access-list log-timer {default|<INTERVAL>}
```

Description

Sets the log timer interval for all ACEs that have the `log` parameter configured.

Command context

```
config
```

Parameters

default

Resets the log timer to its default 300 seconds.

<INTERVAL>

Specifies the log timer interval in seconds. Range: 30 to 300.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- ACL logging keeps a log of the number of packets matching this ACE. Works with both `permit` and `deny` actions. Works with ACLs applied on ingress or egress, except for control plane.
 - The first packet that matches an ACE with the `log` parameter within an ACL log timer window (configured with the `access-list log-timer` command) has its header contents extracted and sent to the configured logging destination, such as the console and syslog server. Each time the ACL log timer expires, a summary of all ACEs with `log` configured are sent to the logging destination. This capability allows throttling of logging ACL hits.
 - If no further log messages are generated in the wait-period, the switch suspends the timer and resets itself to log as soon as a new match occurs.
 - When using multiple ACL types (IPv4, IPv6, or MAC) with logging on the same interface, the first packet that matches an ACE with the `log` option is logged. Any packets, matching other ACL types, do not create a log until the log-timer wait-period is over. At the end of the wait-period, a summary log is made of all the ACLs that were matched, regardless of type.
-
- **NOTE:** Remarkd ACL traffic may lose logging information when a QoS action or a classifier policy with remark is enabled. A classifier policy with remark takes precedence over QoS actions and QoS actions takes precedence over ACL remarkd traffic.
-
- You may see a minor discrepancy between the ACL logging statistics and the hit counts statistics due to the time required to record the log message.

Examples



NOTE: Although these examples use debug logging, you can alternatively use event logging.

Enabling debug logging for the ACL logging module:

```
switch# debug acl log severity info
switch# show debug
-----
module sub_module severity vlan port ip mac instance vrf
-----
acl acl_log info -----
```

Setting the debug destination to console with the minimum security level of info:

```
switch# debug destination console severity info
switch# show debug destination
-----
show debug destination
-----
CONSOLE:info
```

Setting the access list log-timer to 30 seconds:

```
switch(config)# access-list log-timer 30
switch(config)# do show access-list log-timer
ACL log timer length (frequency): 30 seconds
```

Creating an IPv4 ACL with one entry with the log parameter:

```
switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# deny icmp 1.1.1.1 1.1.1.2 log
switch(config-acl-ip)# do show access-list
```

Type	Name		
Sequence	Comment	Action	L3 Protocol
	Source IP Address	Source L4 Port(s)	
	Destination IP Address	Destination L4 Port(s)	
	Additional Parameters		

IPv4	MY_IP_ACL		
10	deny		icmp
	1.1.1.1		
	1.1.1.2		
	Logging: enabled		
	Hit-counts: enabled		

Enabling interface 1/1/1 and applying the ACL:

```
switch(config)# interface 1/1/1
switch(config-if)# apply access-list ip MY_IP_ACL in
switch(config-if)# do show running-config interface 1/1/1
interface 1/1/1
  no shutdown
  apply access-list ip MY_IP_ACL in
  no routing
  vlan access 1
  exit
```

Sending packets that will match the ACE and observe the ACL logging message on the console:

```
2017-10-10T20:13:36.044+00:00 ops-switchd[875]: debug|LOG_INFO|AMM|1/5|ACL|ACL_LOG|
List MY_IP_ACL, seq# 10 denied icmp 1.1.1.1 -> 1.1.1.2 type 8 code 0,
on vlan 1, port 1/1/1, direction in
```

When the access list log-timer expires, the summary message is printed on the console. The number 30 is the number of packets received during the last access list log-timer window.

```
2017-10-10T20:14:06.051+00:00 ops-switchd[875]: debug|LOG_INFO|AMM|1/5|ACL|ACL_LOG|
MY_IP_ACL on 1/1/1 (in): 30 10 deny icmp 1.1.1.1 1.1.1.2 log count
```

Resetting the ACL log timer to the default value:

```
switch(config)# access-list log-timer default
```

access-list mac

Syntax

```
access-list mac <ACL-NAME>
```

```
no access-list mac <ACL-NAME>
```

```
[<SEQUENCE-NUMBER>]
{permit|deny}
{any|<SRC-MAC-ADDRESS>[/<ETHERNET-MASK>]}
{any|<DST-MAC-ADDRESS>[/<ETHERNET-MASK>]}
{any|aarp|appletalk|arp|fcoe|fcoe-init|ip|ipv6|
  ipx-arpa|ipx-non-arpa|is-is|lldp|mpls-multicast|mpls-unicast|q-in-q|
  rbridge|trill|wake-on-lan|<NUMERIC-ETHERTYPE>}
[pcp <PCP-VALUE>] [vlan <VLAN-ID>] [count] [log]

no <SEQUENCE-NUMBER>
```

```
[<SEQUENCE-NUMBER>] comment <TEXT-STRING>
```

```
no <SEQUENCE-NUMBER> comment
```

Description

Creates a MAC Access Control List (ACL). The ACL is made of one or more Access Control Entries (ACEs) ordered and prioritized by sequence numbers. The lowest sequence number is the highest prioritized ACE.

The `no` form of this command deletes the entire ACL, or deletes an ACE identified by sequence number, or deletes only the comment from the ACE identified by sequence number.

Command context

```
config
```

The `access-list mac <ACL-NAME>` command takes you into the named ACL context where you enter the ACEs.

Parameters

<ACL-NAME>

Specifies the name of this ACL.

<SEQUENCE-NUMBER>

Specifies a sequence number for the ACE. Range: 1 to 4294967295.

{permit|deny}

Specifies whether to permit or deny traffic matching this ACE.

comment

Specifies storing the remaining entered text as an ACE comment.

{any|<SRC-MAC-ADDRESS>[/<ETHERNET-MASK>]}]

Specifies the source host MAC address (xxxx.xxxx.xxxx), OUI, or the keyword `any`. You can optionally include the following:

<ETHERNET-MASK> - The address bits to mask (xxxx.xxxx.xxxx).

{any|<DST-MAC-ADDRESS>[/<ETHERNET-MASK>]}]

Specifies the destination host MAC address (xxxx.xxxx.xxxx), OUI, or the keyword `any`. You can optionally include the following:

<ETHERNET-MASK> - The address bits to mask (xxxx.xxxx.xxxx).

{any|arp|appletalk|...|wake-on-lan|<NUMERIC-ETHERTYPE>}

Specifies the protocol encapsulated in the Ethernet frame. The encapsulated protocol is identified by the EtherType Ethernet field. The EtherType is specified in one of the following three ways:

- `any` - any EtherType.
- **<NUMERIC-ETHERTYPE>** - the numerical EtherType protocol number. Range: 0x600 to 0xffff.
- One of these EtherType protocol name keywords:

aarp
appletalk
arp
fcoe
fcoe-init
ip
ipv6
ipx-arpa
ipx-non-arpa
is-is
lldp
mpls-multicast
mpls-unicast
q-in-q
rbridge
trill
wake-on-lan

pcp <PCP-VALUE>

Specifies 802.1Q QoS Priority Code Point value. Range: 0 to 7.

vlan <VID>

Specifies a VLAN ID. The VLAN ID must exist.



NOTE: This parameter cannot be used in any ACL that will be applied to a VLAN.

count

Keeps the hit counts of the number of packets matching this ACE.

log

Keeps a log of the number of packets matching this ACE. Works with both `permit` and `deny` actions. Works with ACLs applied on ingress or egress.

Authority

Administrators or local user group members with execution rights for this command.

Usage

When using multiple ACL types (IPv4, IPv6, or MAC) with logging on the same interface, the first packet that matches an ACE with `log` option is logged. Until the log-timer wait-period is over, any packets matching other ACL types do not create a log. At the end of the wait-period, the switch creates a summary log all the ACLs that were matched, regardless of type.

Examples

Creating a MAC ACL with four entries:

```
switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-ip)# 10 permit 1122.3344.5566/ffff.ffff.0000 any ipv6
switch(config-acl-ip)# 20 permit aaaa.bbbb.cccc 1111.2222.3333 any pcp 4
switch(config-acl-ip)# 30 permit any any appletalk vlan 40
switch(config-acl-ip)# 40 deny any any any count
switch(config-acl-ip)# exit
```

```

switch(config)# do show access-list
Type          Name
Sequence      Comment
              Action
              Source MAC Address
              Destination MAC Address
              Additional Parameters
-----
MAC           MY_MAC_ACL
10 permit
1122.3344.5566/ffff.ffff.0000
any
20 permit
aaaa.bbbb.cccc
1111.2222.3333
QoS Priority Code Point: 4
30 permit
any
any
VLAN: 40
40 deny
any
any
Hit-counts: enabled
EtherType
ipv6
any
appletalk
any

```

Adding a comment to an existing MAC ACE:

```

switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-ip)# 30 comment Permit all vlan-40 tagged Appletalk traffic
switch(config-acl-ip)# exit

```

```

switch(config)# do show access-list
Type          Name
Sequence      Comment
              Action
              Source MAC Address
              Destination MAC Address
              Additional Parameters
-----
MAC           MY_MAC_ACL
10 permit
1122.3344.5566/ffff.ffff.0000
any
20 permit
aaaa.bbbb.cccc
1111.2222.3333
QoS Priority Code Point: 4
30 Permit all vlan-40 tagged Appletalk traffic
permit
any
any
VLAN: 40
40 deny
any
any
Hit-counts: enabled
EtherType
ipv6
any
appletalk
any

```

Removing a comment from an existing MAC ACE:

```

switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-mac)# no 30 comment
switch(config-acl-mac)# exit

```

```

switch(config)# do show access-list
Type      Name
Sequence  Comment
          Action                      EtherType
          Source MAC Address
          Destination MAC Address
          Additional Parameters
-----
MAC       MY_MAC_ACL
10 permit 1122.3344.5566/ffff.ffff.0000  ipv6
    any
20 permit aaaa.bbbb.cccc                any
    1111.2222.3333
    QoS Priority Code Point: 4
30 permit                                appletalk
    any
    any
    VLAN: 1
40 deny                                       any
    any
    any
    Hit-counts: enabled

```

Adding an ACE to an existing MAC ACL:

```

switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-ip)# 35 permit any aabb.cc11.1234 0xffee
switch(config-acl-ip)# exit

```

```

switch(config)# do show access-list
Type      Name
Sequence  Comment
          Action                      EtherType
          Source MAC Address
          Destination MAC Address
          Additional Parameters
-----
MAC       MY_MAC_ACL
10 permit 1122.3344.5566/ffff.ffff.0000  ipv6
    any
20 permit aaaa.bbbb.cccc                any
    1111.2222.3333
    QoS Priority Code Point: 4
30 permit                                appletalk
    any
    any
    VLAN: 1
35 permit                                0xffee
    any
    aabb.cc11.1234
40 deny                                       any
    any
    any
    Hit-counts: enabled

```

Replacing an ACE in an existing MAC ACL:

```

switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-ip)# 35 permit any aabb.cc11.1234 0xeeee
switch(config-acl-ip)# exit

```



```

switch(config)# do show access-list
Type          Name
Sequence      Comment
              Action
              Source MAC Address
              Destination MAC Address
              Additional Parameters
-----
MAC           MY_MAC_ACL
10 permit
1122.3344.5566/ffff.ffff.0000
any
20 permit
aaaa.bbbb.cccc
1111.2222.3333
QoS Priority Code Point: 4
30 permit
any
any
VLAN: 1
35 permit
any
aabb.cc11.1234
40 deny
any
any
Hit-counts: enabled

```

Removing an ACE from an MAC ACL:

```

switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-ip)# no 35
switch(config-acl-ip)# exit

switch(config)# do show access-list
Type          Name
Sequence      Comment
              Action
              Source MAC Address
              Destination MAC Address
              Additional Parameters
-----
MAC           MY_MAC_ACL
10 permit
1122.3344.5566/ffff.ffff.0000
any
20 permit
aaaa.bbbb.cccc
1111.2222.3333
QoS Priority Code Point: 4
30 permit
any
any
VLAN: 1
40 deny
any
any
Hit-counts: enabled

```

Removing a MAC ACL:

```

switch(config)# no access-list mac MY_MAC_ACL

```

```

switch(config)# do show access-list
Type          Name
Sequence      Comment
              Action
              Source MAC Address
              Destination MAC Address
              Additional Parameters
-----
MAC           MY_MAC_ACL2
1 permit      1122.3344.5566/ffff.ffff.0000    ipv6
  any
2 permit      aaaa.bbbb.cccc                    any
  1111.2222.3333
  QoS Priority Code Point: 4
3 Permit all vlan-40 tagged Appletalk traffic
  permit      appletalk
  any
  any
  VLAN: 1
4 deny        any
  any
  any
  Hit-counts: enabled

```

access-list resequence

Syntax

```
access-list {ip|ipv6|mac} <ACL-NAME> resequence <STARTING-SEQUENCE-NUMBER> <INCREMENT>
```

Description

Resequences the ACE sequence numbers in an ACL.

Command context

config

Parameters

{ip|ipv6|mac}

Specifies the ACL type.

<ACL-NAME>

Specifies the ACL name.

<STARTING-SEQUENCE-NUMBER>

Specifies the starting sequence number.

<INCREMENT>

Specifies the sequence number increment.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Resequencing an IPv4 ACL to start at 1 with an increment of 1:

```
switch(config)# access-list ip MY_IP_ACL resequence 1 1
switch(config-acl-ip)# exit
```

```
switch(config)# do show access-list
```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address	Destination L4 Port(s)	Additional Parameters
IPv4	MY_IP_ACL	1	permit	any	udp	172.16.1.0/255.255.255.0				
		2	permit	any	tcp	172.16.2.0/255.255.0.0	> 1023			
		3	permit	any	tcp	172.26.1.0/255.255.255.0				
				dscp: AF11						
				ack						
				syn						
		4	deny	any	any					
				any						
				Hit-counts: enabled						

Resequencing an IPv6 ACL to start at 1 with an increment of 1:

```
switch(config)# access-list ipv6 MY_IPV6_ACL resequence 1 1
switch(config-acl-ip)# exit
```

```
switch(config)# do show access-list
```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address	Destination L4 Port(s)	Additional Parameters
IPv6	MY_IPV6_ACL	1	permit	any	udp	2001::1/64				
		2	Permit all TCP ephemeral ports	permit	tcp	2001:2001::2:1	> 1023			
		3	permit	any	tcp	2001:2011::1/64				
		4	deny	any	any					
				any						
				Hit-counts: enabled						

Resequencing a MAC ACL to start at 1 with an increment of 1:

```
switch(config)# access-list mac MY_MAC_ACL resequence 1 1
switch(config-acl-mac)# exit
```

```
switch(config)# do show access-list
```

Type	Name		
Sequence	Comment		
	Action		EtherType
	Source MAC Address		
	Destination MAC Address		
	Additional Parameters		
MAC	MY_MAC_ACL		
1	permit		ipv6
	1122.3344.5566/ffff.ffff.0000		
	any		
2	permit		any
	aaaa.bbbb.cccc		
	1111.2222.3333		
	QoS Priority Code Point: 4		
3	Permit all vlan-40 tagged Appletalk traffic		
	permit		appletalk
	any		
	any		
	VLAN: 1		
4	deny		any
	any		
	any		
	Hit-counts: enabled		

access-list reset

Syntax

```
access-list {all|ip <ACL-NAME>|ipv6 <ACL-NAME>|mac <ACL-NAME>} reset
```

Description

Changes the user-specified ACL configuration to match the active ACL configuration. Use this command when a discrepancy exists between what the user configured and what is active and accepted by the system.

Command context

```
config
```

Parameters

```
all|ip <ACL-NAME>|ipv6 <ACL-NAME>|mac <ACL-NAME>
```

Specifies **one** of the following:

- a reset of all ACLs.
- a reset of a named IPv4 ACL.
- a reset of a named IPv6 ACL.
- a reset of a named MAC ACL.

Authority

Administrators or local user group members with execution rights for this command.

Usage

The output of the `show access-list` command displays the active configuration of the product. The active configuration is the ACLs that have been configured and accepted by the system. The output of the `show access-list` command with the `configuration` parameter, displays the ACLs that have been configured.

The output of this command may not be the same as what was programmed in hardware or what is active on the product.

If the active ACLs and user-configured ACLs are not the same, a warning message is displayed in the output of the show command. Modify the user-configured ACL until the warning message is no longer displayed or run the `access-list reset` command to change the user-specified configuration to match the active configuration.

Examples

Apply an ACL with TCP acknowledgments (ACKs) on egress, which is unsupported by hardware:

```
switch(config-acl)# 10 permit tcp 172.16.2.0/16 any ack
```

Displaying the user-specified configuration:

```
switch(config)# do show access-list commands
! access-list ip TEST_ACL user configuration does not match active configuration.
! run 'access-list TYPE NAME reset' to reset access-list to match active configuration.
access-list ip TEST_ACL
! access-list ip TEST_ACL user configuration does not match active configuration.
! run 'access-list TYPE NAME reset' to reset access-list to match active configuration.
interface 1/1/1
    apply access-list ip TEST_ACL out

switch(config)# do show access-list commands configuration
! access-list ip TEST_ACL user configuration does not match active configuration.
! run 'access-list TYPE NAME reset' to reset access-list to match active configuration.
access-list ip TEST_ACL
    10 permit tcp 172.16.2.0/255.255.0.0 any ack
! access-list ip TEST_ACL user configuration does not match active configuration.
! run 'access-list TYPE NAME reset' to reset access-list to match active configuration.
interface 1/1/1
    apply access-list ip TEST_ACL out

switch(config)# do show access-list
Type      Name
Sequence Comment
          Action
          Source IP Address
          Destination IP Address
          Additional Parameters
-----
% Warning: TEST_ACL user configuration does not match active configuration.
%          run 'access-list TYPE NAME reset' to reset access-list to match active configuration.
IPv4      TEST_ACL

switch(config)# do show access-list configuration
Type      Name
Sequence Comment
          Action
          Source IP Address
          Destination IP Address
          Additional Parameters
-----
% Warning: TEST_ACL user configuration does not match active configuration.
%          run 'access-list TYPE NAME reset' to reset access-list to match active configuration.
IPv4      TEST_ACL
          10
          permit
          172.16.2.0/255.255.0.0
          any
          ack
          tcp
```

Resetting the user-specified configuration to match the active configuration.

```
switch(config)# access-list ip TEST_ACL reset
```

Displaying the updated user-specified configuration.

```

switch(config)# do show access-list commands
access-list ip TEST_ACL
interface 1/1/1
    apply access-list ip TEST_ACL out

switch(config)# do show access-list commands configuration
access-list ip TEST_ACL
interface 1/1/1
    apply access-list ip TEST_ACL out

switch(config)# do show access-list
Type          Name
Sequence      Comment
              Action                    L3 Protocol
              Source IP Address         Source L4 Port(s)
              Destination IP Address     Destination L4 Port(s)
              Additional Parameters
-----
IPv4          TEST_ACL

switch(config)# do show access-list configuration
Type          Name
Sequence      Comment
              Action                    L3 Protocol
              Source IP Address         Source L4 Port(s)
              Destination IP Address     Destination L4 Port(s)
              Additional Parameters
-----
IPv4          TEST_ACL

```

apply access-list control-plane

Syntax

```

apply access-list {ip|ipv6} <ACL-NAME> control-plane vrf <VRF-NAME>

no apply access-list {ip|ipv6} <ACL-NAME> control-plane vrf <VRF-NAME>

```

Description

Applies an ACL to the specified VRF.

The **no** form of this command removes application of the ACL from the specified VRF.

Command context

config

Parameters

ip|ipv6

Specifies the ACL type: **ip** for IPv4, or **ipv6** for IPv6.

<ACL-NAME>

Specifies the ACL name.

vrf <VRF-NAME>

Specifies the VRF name.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Only one ACL per type (`ip`, or `ipv6`) may be applied to a control plane VRF at a time. Therefore, using the `apply access-list control-plane` command on a VRF with an already-applied ACL of the same type, will replace the applied ACL.

Examples

Applying `My_ip_ACL` to control plane traffic on the default VRF:

```
switch(config)# apply access-list ip My_ip_ACL control-plane vrf default
```

Replacing `My_ip_ACL` with `My_Replacement_ACL` on the default VRF:

```
switch(config)# apply access-list ip My_Replacement_ACL control-plane vrf default
```

Remove (unapply) the `My_Replacement_ACL` from the default VRF. Any other interfaces or VLANs with `My_Replacement_ACL` applied are unaffected.

```
switch(config)# no apply access-list ip My_Replacement_ACL control-plane vrf default
```

apply access-list (to interface or LAG)

Syntax

```
apply access-list {ip|ipv6|mac} <ACL-NAME> {in|out}
```

```
no apply access-list {ip|ipv6|mac} <ACL-NAME> {in|out}
```

Description

Applies an ACL to the interface (Individual front plane port) or Link Aggregation Group (LAG) identified by the current interface or LAG context.

The `no` form of this command removes application of the ACL from the current interface or LAG identified by the current interface or LAG context.

Command context

```
config-if
```

```
config-lag-if
```

Parameters

ip|ipv6|mac

Specifies the ACL type: `ip` for IPv4, `ipv6` for IPv6, or `mac` for MAC ACL.

<ACL-NAME>

Specifies the ACL name.

in|out

Selects the traffic direction on which the ACL is applied.

- `in` selects the inbound (ingress) traffic direction.
- `out` selects the outbound (egress) traffic direction.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- Each ACL of a given type can be applied to the same interface or LAG once in each direction. Therefore, using the `apply access-list` command on an interface or LAG with an already-applied ACL of the same type and direction will replace the currently applied ACL.
- An ACL can be applied to an individual front plane port or to a Link Aggregation Group (LAG).
- A port that is a member of a LAG with an applied ACL cannot have a different ACL applied to that member port.
- When the port membership of a LAG with an applied ACL is changed, the LAG ACL is automatically applied or removed from that port depending on the modification type.

Examples

Applying My_IP_ACL to ingress traffic on interface range 1/1/10 to 1/1/12:

```
switch(config)# int 1/1/10-1/1/12
switch(config-if-<1/1/10-1/1/12>)# apply access-list ip My_IP_ACL in
switch(config-if-<1/1/10-1/1/12>)# exit
```

Applying MY_IP_ACL to ingress traffic on LAG 100 and apply to egress traffic on interface 1/1/2:

```
switch(config)# interface lag 100
switch(config-lag-if)# apply access-list ip MY_IP_ACL in
switch(config-lag-if)# exit

switch(config)# interface 1/1/2
switch(config-if)# apply access-list ip MY_IP_ACL out
switch(config-if)# exit
switch(config)#
```

Applying MY_IPV6_ACL to ingress traffic on interface 1/1/1 and apply to ingress traffic on LAG 100:

```
switch(config)# interface 1/1/1
switch(config-if)# apply access-list ipv6 MY_IPV6_ACL in
switch(config-if)# exit

switch(config)# interface lag 100
switch(config-lag-if)# apply access-list ipv6 MY_IPV6_ACL in
switch(config-lag-if)# exit
switch(config)#
```

Applying MY_MAC_ACL to ingress traffic on interface 1/1/1 and apply to ingress traffic on interface 1/1/2:

```
switch(config)# interface 1/1/1
switch(config-if)# apply access-list mac MY_MAC_ACL in
switch(config-if)# exit

switch(config)# interface 1/1/2
switch(config-if)# apply access-list mac MY_MAC_ACL in
switch(config-if)# exit
switch(config)#
```

Replacing MY_IP_ACL with MY_REPLACEMENT_ACL on interface 1/1/2:

```
switch(config)# interface 1/1/2
switch(config-if)# apply access-list ip MY_REPLACEMENT_ACL out
```



```
switch(config-if)# exit  
switch(config)#
```

Unapplying MY_REPLACEMENT_ACL from interface 1/1/2 (out):

```
switch(config)# interface 1/1/2  
switch(config-if)# no apply access-list ip MY_REPLACEMENT_ACL out  
switch(config-if)# exit  
switch(config)#
```

apply access-list (to interface VLAN)

Syntax

```
apply access-list {ip|ipv6} <ACL-NAME> {routed-in|routed-out}
```

```
no apply access-list {ip|ipv6} <ACL-NAME> {routed-in|routed-out}
```

Description

Applies an ACL to the interface VLAN (or range of interface VLANs) identified by the current interface VLAN context. Using the `apply access-list` command on an interface VLAN interface with an already-applied ACL of the same direction and type will replace the currently-applied ACL.

The `no` form of this command removes application of the ACL from the interface VLAN (or range of interface VLANs) identified by the current interface VLAN context.

Command context

```
config-if-vlan
```

Parameters

ip|ipv6

Specifies the ACL type: `ip` for IPv4, `ipv6` for IPv6.

<ACL-NAME>

Specifies the ACL name.

routed-in

Selects the routed inbound (routed ingress) traffic direction.

routed-out

Selects the routed outbound (routed egress) traffic direction.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- Each ACL of a given type can be applied to the same interface VLAN once in each direction. Therefore, using the `apply access-list` command on an interface VLAN with an already-applied ACL of the same direction and type, will replace the applied ACL.
- When an ACL is applied to an interface VLAN, it will create hardware entries on all stack members regardless of whether an interface VLAN member exists on any specific stack member.

Examples

Creating an IPv4 ACL and applying it to routed ingress traffic on interface VLAN vlan100:

```
switch(config)# access-list ip test
switch(config-acl-ip)# 10 permit any 1.1.1.2 2.2.2.2 count
switch(config-acl-ip)# 20 permit any 1.1.1.2 2.2.2.1 count
switch(config-acl-ip)# 30 permit any 2.2.2.2 1.1.1.2 count
switch(config-acl-ip)# 40 permit any 2.2.2.2 1.1.1.1 count
switch(config-acl-ip)# 50 permit any any any count
switch(config-acl-ip)# exit
switch(config)#
switch(config)# interface vlan100
switch(config-if-vlan)# apply access-list ip test routed-in
```

Applying My_ip_ACL to routed ingress traffic on interface VLAN 10:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# apply access-list ip My_ip_ACL routed-in
```

Applying My_ipv6_ACL to routed ingress traffic on interface VLAN 10:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# apply access-list ipv6 My_ip_ACL routed-in
```

Applying My_ip_ACL to routed ingress traffic on interface VLANs 20 to 25:

```
switch(config)# interface vlan 20-25
switch(config-if-vlan-<20-25># apply access-list ip My_ip_ACL routed-in
```

Replacing My_ipv6_ACL with My_Replacement_ACL on interface VLAN 10 (following the above examples):

```
switch(config)# interface vlan 10
switch(config-if-vlan)# apply access-list ipv6 My_Replacement_ACL routed-in
```

Removing (unapplying) My_Replacement_ACL on interface VLAN 10. Any other interfaces or VLANs with My_Replacement_ACL applied are not affected:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# no apply access-list ipv6 My_Replacement_ACL routed-in
```

Removing (unapplying) My_ip_ACL on interface VLANs 20 to 25. Any other interfaces or VLANs with My_ip_ACL applied are not affected:

```
switch(config)# interface vlan 20-25
switch(config-if-vlan-<20-25># no apply access-list ip My_ip_ACL routed-in
```

Applying My_ip_ACL to routed egress traffic on interface VLAN 30:

```
switch(config)# interface vlan 30
switch(config-if-vlan)# apply access-list ip My_ip_ACL routed-out
```

Applying My_ip_ACL to routed egress traffic on interface VLANs 40 to 50:

```
switch(config)# interface vlan 40-50
switch(config-if-vlan-<40-50># apply access-list ip My_ip_ACL routed-out
```

apply access-list (to VLAN)

Syntax

```
apply access-list {ip|ipv6|mac} <ACL-NAME> {in|out}

no apply access-list {ip|ipv6|mac} <ACL-NAME> {in|out}
```

Description

Applies an ACL to the VLAN identified by the current VLAN context.

The `no` form of this command removes application of the ACL from the VLAN identified by the current VLAN context.

Command context

```
config-vlan
```

Parameters

ip|ipv6|mac

Specifies the ACL type: `ip` for IPv4, `ipv6` for IPv6, or `mac` for MAC ACL.

<ACL-NAME>

Specifies the ACL name.

in

Selects the inbound (ingress) traffic direction.

out

Selects the outbound (egress) traffic direction.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- Each ACL of a given type can be applied to the same VLAN once in each direction. Therefore, using the `apply access-list` command on a VLAN with an already-applied ACL of the same direction and type, will replace the applied ACL.
- When an ACL is applied to a VLAN, it will create hardware entries on all stack members regardless of whether a VLAN member exists on any specific stack member.

Examples

Applying `My_ip_ACL` to ingress traffic on VLAN range 20 to 25:

```
switch(config)# vlan 20-25
switch(config-vlan-<20-25>)# apply access-list ip My_ip_ACL in
```

Applying `My_ip_ACL` to egress traffic on VLAN range 40 to 50:

```
switch(config)# vlan 40-50
switch(config-vlan-<40-50>)# apply access-list ip My_ip_ACL out
```

Applying `My_ip_ACL` to ingress traffic on VLAN 10:

```
switch(config)# vlan 10  
switch(config-vlan-10)# apply access-list ip My_ip_ACL in
```

Applying My_ipv6_ACL to ingress traffic on VLAN 10:

```
switch(config)# vlan 10  
switch(config-vlan-10)# apply access-list ipv6 My_ipv6_ACL in
```

Applying My_mac_ACL to ingress traffic on VLAN 10:

```
switch(config)# vlan 10  
switch(config-vlan-10)# apply access-list mac My_mac_ACL in
```

Replacing My_ipv6_ACL with My_Replacement_ACL on VLAN 10 (following the preceding examples):

```
switch(config)# vlan 10  
switch(config-vlan-10)# apply access-list ipv6 My_Replacement_ACL in
```

Removing (unapplying) several ACLs on VLAN 10:

```
switch(config)# vlan 10  
switch(config-vlan-10)# no apply access-list ipv6 My_Replacement_ACL in  
switch(config-vlan-10)# no apply access-list mac My_mac_ACL in
```

clear access-list hitcounts

Syntax

```
clear access-list hitcounts { all | [{ip|ipv6|mac} <ACL-NAME>] [interface <IF-NAME>|  
vlan <VLAN-ID>] [in|out|routed-in|routed-out] }
```

Description

Clears the hit counts for ACLs with ACEs that include the `count` keyword.

Command context

Operator (>) or Manager (#)

Parameters

all

Selects all ACLs.

ip|ipv6|mac

Specifies the ACL type: `ip` for IPv4, `ipv6` for IPv6, or `mac` for MAC.

<ACL-NAME>

Specifies the ACL name.

interface <IF-NAME>

Specifies the interface name (port or LAG).

vlan <VLAN-ID>

Specifies the VLAN.

in|out

Selects the traffic direction on which the ACL is applied.

- `in` selects the inbound (ingress) traffic direction.
- `out` selects the outbound (egress) traffic direction.

routed-in|routed-out

Selects the routed traffic direction on which the ACL is applied.



NOTE: This is only available for IPv4 and IPv6 ACLs applied to interface VLANs.

- `routed-in` selects the routed inbound (routed ingress) traffic direction.
- `routed-out` selects the routed outbound (routed egress) traffic direction.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Clearing the hit counts for My_ip_ACL applied to port 1/1/2 (egress):

```
switch# clear access-list hitcounts ip My_ip_ACL interface 1/1/2 out
```

Clearing the hit counts for My_ip_ACL applied to VLAN 10 (ingress):

```
switch# clear access-list hitcounts ip My_ip_ACL vlan 10 in
```

Clearing the hit counts for all ACLs:

```
switch# clear access-list hitcounts all
```

clear access-list hitcounts control-plane

Syntax

```
clear access-list hitcounts [{ip|ipv6} <ACL-NAME>] control-plane vrf <VRF-NAME>
```

Description

Clears the hit counts for ACLs applied to the specified Control Plane VRF.

Command context

Operator (>) or Manager (#)

Parameters

ip|ipv6

Specifies the ACL type: `ip` for IPv4, or `ipv6` for IPv6.

<ACL-NAME>

Specifies the ACL name.

vrf <VRF-NAME>

Specifies the VRF name.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Clearing the hit counts for an IPv4 ACL applied to the Control Plane mgmt VRFs:

```
switch# clear access-list hitcounts ip My_ipv4_ACL control-plane vrf mgmt
```

Clearing the hit counts for an IPv6 ACL applied to all Control Plane VRFs:

```
switch# clear access-list hitcounts ipv6 My_ipv6_ACL control-plane
```

object-group address resequence

Syntax

```
object-group {ip|ipv6} address <OBJECT-GROUP-NAME> resequence <STARTING-SEQUENCE-NUMBER> <INCREMENT>
```

Description

Reorders the sequence numbers in an address object group.

Command context

config

Parameters

ip|ipv6

Specifies the object group IP address type, either `ip` or `ipv6`.

<OBJECT-GROUP-NAME>

Specifies the address object group name.

<STARTING-SEQUENCE-NUMBER>

Specifies the starting sequence number.

<INCREMENT>

Specifies the sequence number increment.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Resequencing address object group `my_ipv4_addr_group` to use sequence numbers 5, 10, 15 and so on:

```
switch(config)# object-group address my_ipv4_addr_group resequence 5 5  
switch(config)#
```

object-group address reset

Syntax

```
object-group {ip|ipv6} address <OBJECT-GROUP-NAME> reset
```

Description

Resets the user configuration back to the active configuration. This command takes immediate effect, it is not saved in the user configuration. Use this command if misconfiguration of an address object group has occurred.

Command context

config

Parameters

ip|ipv6

Specifies the object group IP address type, either `ip` or `ipv6`.

<OBJECT-GROUP-NAME>

Specifies the address object group name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Resetting IPv4 address object group `my_ipv4_group`:

```
switch(config)# object-group ip address my_ip_group reset
switch(config)#
```

Resetting IPv6 address object group `my_ipv6_group`:

```
switch(config)# object-group ipv6 address my_ipv6_group reset
switch(config)#
```

object-group all reset

Syntax

```
object-group all reset
```

Description

Resets the user configuration back to the active configuration for all object types (address and port). This command takes immediate effect, it is not saved in the user configuration. Use this command if misconfiguration of address object groups and port object groups has occurred. Individual address and port object groups can be reset respectively with the `object-group address reset` and `object-group port reset` commands.

Command context

config

Authority

Administrators or local user group members with execution rights for this command.

Examples

Resetting the user configuration for all object types (address and port):

```
switch(config)# object-group all reset
switch(config)#
```

object-group ip address

Syntax

Syntax to create an IPv4 address object group and enter its context:

```
object-group ip address <OBJECT-GROUP-NAME>
```

```
no object-group ip address <OBJECT-GROUP-NAME>
```

Syntax (within the address object-group context) for creating or removing IPv4 address entries:

```
[ <SEQUENCE-NUMBER>] <IP-ADDRESS> [ / { <PREFIX-LENGTH> | <SUBNET-MASK> } ]
```

```
no <SEQUENCE-NUMBER>
```

Description

Creates an IPv4 address object group comprised of one or more address entries. Address groups are used solely as a shorthand way of specifying groups of addresses in the ACEs that make up ACLs. IPv4 address groups can be used only in the `access-list ip` command. Entering `object-group ip address` with an existing address group name, enables you to modify an existing address group.

The `no` form of this command deletes the entire address group or deletes a particular address group entry identified by sequence number.

Command context

```
config
```

The `object-group ip address` command takes you into the named address group context (with prompt `switch(config-addrgroup-ip)#`) where you enter the addresses.

Parameters

<OBJECT-GROUP-NAME>

Specifies the address object group name.

<SEQUENCE-NUMBER>

Specifies a sequence number for the address entry. Range: 1 to 4294967295. When omitted, a sequence number 10 larger than the current highest sequence number is auto-assigned. Default auto-assigned sequence numbers are 10, 20, 30, and so on.

<IP-ADDRESS> [/ { <PREFIX-LENGTH> | <SUBNET-MASK> }]

Specifies the IPv4 address.

- **<IP-ADDRESS>** - specifies the IPv4 host address.
 - **<PREFIX-LENGTH>** - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32.
 - **<SUBNET-MASK>** - specifies the address bits to mask (dotted decimal notation).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating an IPv4 address group with two entries:

```
switch(config)# object-group ip address my_ipv4_addr_group
switch(config-addrgroup-ip)# 10 192.168.0.1
switch(config-addrgroup-ip)# 20 192.168.0.2
switch(config-addrgroup-ip)# exit
switch(config)# show object-group
Type          Name
Sequence L4 Port(s)/IP Address
-----
IPv4         my_ipv4_addr_group
           10 192.168.0.1
           20 192.168.0.2
```

Adding an entry to an existing IPv4 address group:

```
switch(config)# object-group ip address my_ipv4_addr_group
switch(config-addrgroup-ip)# 30 192.168.0.3
switch(config-addrgroup-ip)# exit
switch(config)# show object-group
Type          Name
Sequence L4 Port(s)/IP Address
-----
IPv4         my_ipv4_addr_group
           10 192.168.0.1
           20 192.168.0.2
           30 192.168.0.3
```

Removing an entry (20) from an existing IPv4 address group:

```
switch(config)# object-group ip address my_ipv4_addr_group
switch(config-addrgroup-ip)# no 20
switch(config-addrgroup-ip)# exit
switch(config)# show object-group
Type          Name
Sequence L4 Port(s)/IP Address
-----
IPv4         my_ipv4_addr_group
           10 192.168.0.1
           30 192.168.0.3
```

Removing an IPv4 address group:

```
switch(config)# no object-group ip address my_ipv4_addr_group
switch(config)# show object-group
No object group found.
```

object-group ipv6 address

Syntax

Syntax to create an IPv6 address object group and enter its context:

```
object-group ipv6 address <OBJECT-GROUP-NAME>
```

```
no object-group ipv6 address <OBJECT-GROUP-NAME>
```

Syntax (within the address object-group context) for creating or removing IPv6 address entries:

```
[ <SEQUENCE-NUMBER> ] <IP-ADDRESS> [ / { <PREFIX-LENGTH> | <SUBNET-MASK> } ]
```

```
no <SEQUENCE-NUMBER>
```

Description

Creates an IPv6 address object group comprised of one or more address entries. Address groups are used solely as a shorthand way of specifying groups of addresses in the ACEs that make up ACLs. IPv6 address groups can be used only in the `access-list ipv6` command. Entering `object-group ipv6 address` with an existing address group name, enables you to modify an existing address group.

The `no` form of this command deletes the entire address group or deletes a particular address group entry identified by sequence number.

Command context

`config`

The `object-group ipv6 address` command takes you into the named address group context (with prompt `switch(config-addrgroup-ipv6)#`) where you enter the addresses.

Parameters

<OBJECT-GROUP-NAME>

Specifies the address object group name.

<SEQUENCE-NUMBER>

Specifies a sequence number for the address entry. Range: 1 to 4294967295. When omitted, a sequence number 10 larger than the current highest sequence number is auto-assigned. Default auto-assigned sequence numbers are 10, 20, 30, and so on.

<IP-ADDRESS> [/ { <PREFIX-LENGTH> | <SUBNET-MASK> }]

Specifies the IPv6 address.

- **<IP-ADDRESS>** - specifies the IPv6 host address.
 - **<PREFIX-LENGTH>** - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 128.
 - **<SUBNET-MASK>** - specifies the address bits to mask (dotted decimal notation).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating an IPv6 address group with two entries:

```
switch(config)# object-group ipv6 address my_ipv6_addr_group
switch(config-addrgroup-ipv6)# 10 1000::1
switch(config-addrgroup-ipv6)# 20 1000::2
switch(config-addrgroup-ipv6)# exit
switch(config)# show object-group
Type          Name
Sequence L4 Port(s)/IP Address
-----
IPv6         my_ipv6_addr_group
           10 1000::1
           20 1000::2
```

Adding an entry to an existing IPv6 address group:

```
switch(config)# object-group ipv6 address my_ipv6_addr_group
switch(config-addrgroup-ipv6)# 30 1000::3
switch(config-addrgroup-ipv6)# exit
```

```
switch(config)# show object-group
Type          Name
Sequence L4 Port(s)/IP Address
-----
IPv6         my_ipv6_addr_group
           10 1000::1
           20 1000::2
           30 1000::3
```

Removing an entry (20) from an existing IPv6 address group:

```
switch(config)# object-group ipv6 address my_ipv6_addr_group
switch(config-addrgroup-ipv6)# no 20
switch(config-addrgroup-ipv6)# exit
switch(config)# show object-group
Type          Name
Sequence L4 Port(s)/IP Address
-----
IPv6         my_ipv6_addr_group
           10 1000::1
           30 1000::3
```

Removing an IPv6 address group:

```
switch(config)# no object-group ipv6 address my_ipv6_addr_group
switch(config)# show object-group
No object group found.
```

object-group port

Syntax

Syntax to create a Layer 4 port object group and enter its context:

```
object-group port <OBJECT-GROUP-NAME>
```

```
no object-group port <OBJECT-GROUP-NAME>
```

Syntax (within the port object-group context) for creating or removing Layer 4 port entries:

```
[<SEQUENCE-NUMBER>] { {eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT> }
```

```
no <SEQUENCE-NUMBER>
```

Description

Creates a Layer 4 port object group comprised of one or more port entries. Port groups are used solely as a shorthand way of specifying groups of ports in the ACEs that make up ACLs. Layer 4 port groups can be used only in the `access-list ip` and `access-list ipv6` commands. Entering `object-group port` with an existing port group name, enables you to modify an existing port group.

The `no` form of this command deletes the entire port group or deletes a particular port group entry identified by sequence number.

Command context

```
config
```

The `object-group ip port` command takes you into the named port group context (with prompt `switch(config-portgroup)#`) where you specify the ports.

Parameters

<OBJECT-GROUP-NAME>

Specifies the port object group name.

<SEQUENCE-NUMBER>

Specifies a sequence number for the port entry. Range: 1 to 4294967295. When omitted, a sequence number 10 larger than the current highest sequence number is auto-assigned. Default auto-assigned sequence numbers are 10, 20, 30, and so on.

{ {eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT> }

Specifies the port or port range. Port numbers are in the range of 0 to 65535.

- eq <PORT> - specifies the Layer 4 port.
- gt <PORT> - specifies any Layer 4 port greater than the indicated port.
- lt <PORT> - specifies any Layer 4 port less than the indicated port.
- range <MIN-PORT> <MAX-PORT> - specifies the Layer 4 port range.



NOTE: When ACLs using ACEs defined with port groups are applied, the same number of hardware resources are consumed as when the ports are specified directly in the ACEs and not in a group. Keep this in mind when creating port groups that include many ports. Although hardware resource consumption is the same, with or without port groups used, it may not be immediately obvious that some port groups that you have defined, include many ports. It is recommended that you name port groups in a manner that reminds you that a group includes many ports.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating a port group with two entries to cover port 80 plus ports 0 through 50:

```
switch(config)# object-group port my_port_group
switch(config-portgroup)# 10 eq 80
switch(config-portgroup)# 20 range 0 50
switch(config-portgroup)# exit
switch(config)# show object-group
Type          Name
Sequence L4 Port(s)/IP Address
-----
Port        my_port_group
           10 eq 80
           20 range 0 50
```

Adding an entry for ports greater than 65525 (covers ports 65526 through 65535):

```
switch(config)# object-group port my_port_group
switch(config-portgroup)# 30 gt 65525
switch(config-portgroup)# exit
switch(config)# show object-group
Type          Name
Sequence L4 Port(s)/IP Address
-----
Port        my_port_group
           10 eq 80
```

```
20 range 0 50
30 gt 65525
```

Removing an entry (#20) from the port group:

```
switch(config)# object-group port my_port_group
switch(config-portgroup)# no 20
switch(config-portgroup)# exit
switch(config)# show object-group
Type          Name
Sequence L4 Port(s)/IP Address
-----
Port         my_port_group
          10 eq 80
          30 gt 65525
```

Removing the port group:

```
switch(config)# no object-group port my_port_group
switch(config)# show object-group
No object group found.
```

object-group port resequence

Syntax

```
object-group port <OBJECT-GROUP-NAME> resequence <STARTING-SEQUENCE-NUMBER> <INCREMENT>
```

Description

Reorders the sequence numbers in a port object group.

Command context

config

Parameters

<OBJECT-GROUP-NAME>

Specifies the port object group name.

<STARTING-SEQUENCE-NUMBER>

Specifies the starting sequence number.

<INCREMENT>

Specifies the sequence number increment.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Resequencing port object group my_port_group to use sequence numbers 110, 120, 130 and so on:

```
switch(config)# object-group port my_port_group resequence 110 10
switch(config)#
```

object-group port reset

Syntax

```
object-group port <OBJECT-GROUP-NAME> reset
```

Description

Resets the user configuration back to the active configuration. This command takes immediate effect, it is not saved in the user configuration. Use this command if misconfiguration of a port object group has occurred.

Command context

config

Parameters

<OBJECT-GROUP-NAME>

Specifies the port object group name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Resetting port object group my_port_group:

```
switch(config)# object-group port my_port_group reset
switch(config)#
```

show access-list

Syntax

Syntax that filters by ACLs applied to an interface or VLAN:

Syntax that filters by the named ACL:

```
show access-list [ip|ipv6|mac] [<ACL-NAME>]
                  [commands] [configuration] [vsx-peer]
```

Description

Shows information about your defined ACLs and where they have been applied. When `show access-list` is entered without parameters, information for all ACLs is shown. The parameters filter the list of ACLs for which information is shown.

Available filtering includes:

- The content of a specific ACL.
- All ACLs of a specific type.
- All ACLs applied to a specific interface (port or LAG).
- All ACLs applied to a specific VLAN.
- All ACLs applied in a particular direction.
- All IPv4 or IPv6 ACLs applied to interface VLANs (routed in or out).

Command context

Operator (>) or Manager (#)

Parameters

interface <IF-NAME>

Specifies the interface name (port or LAG).

vlan <VLAN-ID>

Specifies the VLAN.

ip|ipv6|mac

Specifies the ACL type: `ip` for IPv4, `ipv6` for IPv6, or `mac` for MAC.

in|out

Selects the traffic direction on which the ACL is applied.

- `in` selects the inbound (ingress) traffic direction.
- `out` selects the outbound (egress) traffic direction.

routed-in|routed-out

Selects the routed traffic direction on which the ACL is applied.



NOTE: This is only available for IPv4 and IPv6 ACLs applied to interface VLANs.

- `routed-in` selects the routed inbound (routed ingress) traffic direction.
- `routed-out` selects the routed outbound (routed egress) traffic direction.

<ACL-NAME>

Specifies the ACL name.

commands

Specifies that the ACL definition is to be shown as the commands and parameters used to create it rather than in tabular form.

configuration

Specifies that the user-configured ACLs be shown as entered, even if the ACLs are not active due to ACE-definition command issues or hardware issues. This parameter is useful if there is a mismatch between the entered configuration and the previous successfully programmed (active) ACLs configuration.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Creating an IPv4 ACL, applying it to an interface VLAN (routed in), and then showing ACL information filtered for that interface VAN:

```
switch(config)# access-list ip test
switch(config-acl-ip)# 10 permit any 1.1.1.2 2.2.2.2 count
switch(config-acl-ip)# 20 permit any 1.1.1.2 2.2.2.1 count
switch(config-acl-ip)# 30 permit any 2.2.2.2 1.1.1.2 count
switch(config-acl-ip)# 40 permit any 2.2.2.2 1.1.1.1 count
switch(config-acl-ip)# 50 permit any any any count
switch(config-acl-ip)# exit
switch(config)#
switch(config)# interface vlan100
switch(config-if-vlan)# apply access-list ip test routed-in
switch(config-if-vlan)# exit
switch(config)# show access-list interface vlan100 ip routed-in
```

Direction

Type	Name
Sequence	Comment
	Ac L3 Protocol
	Source IP Address
	Destination IP Address
	Additional Parameters

Routed Inbound

IPv4	test
10	permit 1.1.1.2 2.2.2.2 Hit-counts: enabled
20	permit 1.1.1.2 2.2.2.1 Hit-counts: enabled
30	permit 2.2.2.2 1.1.1.2 Hit-counts: enabled
40	permit 2.2.2.2 1.1.1.1 Hit-counts: enabled
50	permit any any Hit-counts: enabled

Showing an IPv4 ACL:

```
switch# show access-list ip MY_ACL
```

Type	Name
Sequence	Comment
	Action
	Source IP Address
	Destination IP Address
	Additional Parameters


```

-----
IPv4      MY_ACL
10 permit                udp
    any
    172.16.1.0/255.255.255.0
20 permit                tcp
    172.16.2.0/255.255.0.0
    > 1023
    any
30 permit                tcp
    172.26.1.0//255.255.255.0
    any
    syn
    ack
    dscp 10
40 deny                  any
    any
    any
Hit-counts: enabled
-----

```

Showing an IPv4 ACL as commands:

```

switch# show access-list ip MY_ACL commands
access-list ip MY_ACL
  10 permit udp any 172.16.1.0/255.255.255.0
  20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any
  30 permit tcp 172.26.1.0/255.255.255.0 any syn ack dscp 10
  40 deny any any any count

```

Showing IPv4 ACLs applied to VLAN 10, inbound:

```

switch# show access-list vlan 10 ip in
Type      Name
Sequence  Comment
          Action
          Source IP Address
          Destination IP Address
          Additional Parameters
-----
IPv4      My_ip_ACL
10 permit                udp
    any
    172.16.1.0/255.255.255.0
20 permit                tcp
    172.16.2.0/255.255.0.0
    > 1023
    any
30 permit                tcp
    172.26.1.0//255.255.255.0
    any
    syn
    ack
    dscp 10
40 deny                  any
    any
    any
Hit-counts: enabled
-----

```

Showing IPv6 ACLs applied to LAG 128, inbound:

```

switch# show access-list interface lag128 ipv6 in
Type      Name
Sequence  Comment
          Action
          L3 Protocol

```

	Source IP Address	Source L4 Port(s)
	Destination IP Address	Destination L4 Port(s)
	Additional Parameters	

IPv6	MY_IPV6_ACL	
10	permit	udp
	any	
	2001::1/64	
20	permit	tcp
	2001:2001::2:1/128	> 1023
	any	
30	permit	tcp
	2001:2011::1/64	
40	deny	any
	any	
	any	
	Hit-counts: enabled	

Showing an IPv6 ACL as commands:

```
switch# show access-list ipv6 MY_IPV6_ACL commands
access-list ipv6 MY_IPV6_ACL
  10 permit udp any 2001::1/64
  20 permit tcp 2001:2001::2:1/128 gt 1023 any
  40 deny any any any count
```

Showing a MAC ACL:

switch# show access-list mac MY_MAC_ACL			
Type	Name		
Sequence	Comment		
	Action	EtherType	
	Source MAC Address		
	Destination MAC Address		
	Additional Parameters		

MAC	MY_MAC_ACL		
10	permit	ipv6	
	1122.3344.5566/ffff.ffff.0000		
	any		
20	permit	any	
	aaaa.bbbb.cccc		
	1111.2222.3333		
	QoS Priority Code Point: 4		
30	deny	any	
	any		
	any		
	Hit-counts: enabled		

Showing a MAC ACL as commands:

```
switch# show access-list mac MY_MAC_ACL commands
access-list mac MY_MAC_ACL
  10 permit 1122.3344.5566/ffff.ffff.0000 any ipv6
  20 permit aaaa.bbbb.cccc 1111.2222.3333 any pc 4
  30 deny any any any count
```

show access-list control-plane

Syntax

```
show access-list [ip|ipv6] [<ACL-NAME>] control-plane [vrf <VRF-NAME>]  
[commands] [configuration] [vsx-peer]
```

Description

Shows information about your defined ACLs that have been applied to the Control Plane. When `show access-list control-plane` is entered without parameters, information for all ACLs applied to the Control Plane is shown. The parameters filter the list of ACLs for which information is shown.

Available filtering includes:

- The content of a specific ACL that has been applied to the Control Plane.
- All ACLs of a specific type that have been applied to the Control Plane.
- All ACLs applied to the Control Plane for a specific VRF.

Command context

Operator (>) or Manager (#)

Parameters

ip|ipv6

Specifies the ACL type: `ip` for IPv4, or `ipv6` for IPv6.

<ACL-NAME>

Specifies the ACL name.

vrf <VRF-NAME>

Specifies the VRF name.

[commands]

Specifies that the ACL definition is to be shown as the commands and parameters used to create it rather than in tabular form.

[configuration]

Specifies that the user-configured ACLs be shown as entered, even if the ACLs are not active due to ACE-definition command issues or hardware issues. This parameter is useful if there is a mismatch between the entered configuration and the previous successfully programmed (active) ACLs configuration.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing an IPv4 ACL applied to the Control Plane `mgmt` VRF:

```

switch# show access-list ip My_ipv4_ACL control-plane vrf mgmt
Type      Name
Sequence  Comment
          Action
          Source IP Address
          Destination IP Address
          Additional Parameters
-----
IPv4      My_ipv4_ACL
10 permit                                udp
   any
   172.16.1.0/24
20 permit                                tcp
   172.16.2.0/16
   any                                   > 1023
30 permit                                tcp
   172.26.1.0/24
   any
   syn
   ack
   dscp 10
40 deny                                    any
   any
   any
Hit-counts: enabled
-----

```

Showing an IPv6 ACL applied to the Control Plane mgmt VRF:

```

switch# show access-list ipv6 My_ipv6_ACL control-plane vrf mgmt
Type      Name
Sequence  Comment
          Action
          Source IP Address
          Destination IP Address
          Additional Parameters
-----
IPv6      My_ipv6_ACL
10 permit                                udp
   any
   2001::1/64
20 permit                                tcp
   2001:2001::2:1/128
   any                                   > 1023
30 permit                                tcp
   2001:2011::1/64
40 deny                                    any
   any
   any
Hit-counts: enabled
-----

```

show access-list hitcounts

Syntax

```

show access-list hitcounts { [{ip|ipv6|mac} <ACL-NAME>] [interface <IF-NAME> |
                             vlan <VLAN-ID>] [in|out|routed-in|routed-out] [vsx-peer] }

```

Description

Shows the hit count of the number of times an ACL has matched a packet or frame for ACEs with the `count` keyword. For ACEs without the `count` keyword, a dash is shown in place of a hit count.

Command context

Operator (>) or Manager (#)

Parameters

ip|ipv6|mac

Specifies the ACL type: `ip` for IPv4, `ipv6` for IPv6, or `mac` for MAC.

<ACL-NAME>

Specifies the ACL name.

interface <IF-NAME>

Specifies the interface name (port or LAG).

vlan <VLAN-ID>

Specifies the VLAN.

in|out

Selects the traffic direction on which the ACL is applied.

- `in` selects the inbound (ingress) traffic direction.
- `out` selects the outbound (egress) traffic direction.

routed-in|routed-out

Selects the routed traffic direction on which the ACL is applied.



NOTE: This is only available for IPv4 and IPv6 ACLs applied to interface VLANs.

- `routed-in` selects the routed inbound (routed ingress) traffic direction.
- `routed-out` selects the routed outbound (routed egress) traffic direction.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

- ACL hit counts are aggregated across all:
 - physical interfaces to which the ACL is applied to on ingress,
 - physical interfaces to which the ACL is applied to on egress,
 - VLANs to which the ACL is applied to on ingress.
 - VLANs to which the ACL is applied to on egress.

- Interface VLANs to which the IPv4 or IPv6 ACL is applied on routed ingress
- Interface VLANs to which the IPv4 or IPv6 ACL is applied on routed egress
- If an ACL with an ACE with the `count` keyword is applied to multiple physical interfaces or VLANs, the hit counts are aggregated. There is one aggregation for physical interfaces and another for VLANs.
- Accumulated hit counts for an applied ACL are cleared upon any modification of the ACL.

Examples

Showing the hit counts for My_ip_ACL applied to port 1/1/2:

```
switch# show access-list hitcounts ip My_ip_ACL interface 1/1/2
Statistics for ACL My_ip_ACL (ipv4):
interface 1/1/2* (out):
    Hit Count  Configuration
    - 10 permit udp any 172.16.1.0/24
    - 20 permit tcp 172.16.2.0/16 gt 1023 any
    - 30 permit tcp 172.26.1.0/24 any syn ack dscp 10
    0 40 deny any any any count
* access-list statistics are shared among each combination of context type
(interface, VLAN, VRF) and direction (in, out, routed-in, routed-out,
control-plane). Use 'access-list TYPE NAME copy' to create a new access-list
for separate statistics.
```

Showing the hit counts for My_ip_ACL applied to VLAN 10:

```
switch# show access-list hitcounts ip My_ip_ACL vlan 10
Statistics for ACL My_ip_ACL (ipv4):
vlan 10* (in):
    Hit Count  Configuration
    - 10 permit udp any 172.16.1.0/24
    - 20 permit tcp 172.16.2.0/16 gt 1023 any
    - 30 permit tcp 172.26.1.0/24 any syn ack dscp 10
    0 40 deny any any any count
* access-list statistics are shared among each combination of context type
(interface, VLAN, VRF) and direction (in, out, routed-in, routed-out,
control-plane). Use 'access-list TYPE NAME copy' to create a new access-list
for separate statistics.
```

show access-list hitcounts control-plane

Syntax

```
show access-list hitcounts [{ip|ipv6} <ACL-NAME>] control-plane vrf <VRF-NAME> [vsx-peer]
```

Description

Shows the hit count of the number of times an ACL (applied to the Control Plane) has matched a packet for ACEs with the `count` keyword. For ACEs without the `count` keyword, a dash is shown in place of a hit count.

Command context

Operator (>) or Manager (#)

Parameters

ip|ipv6

Specifies the ACL type: `ip` for IPv4, or `ipv6` for IPv6.

<ACL-NAME>

Specifies the ACL name.

vrf <VRF-NAME>

Specifies the VRF name.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

- ACL hit counts are aggregated across all VRFs to which the ACL is applied to on ingress.
- Accumulated hit counts for an applied ACL are cleared upon any modification of the ACL.

Examples

Showing the hit counts for an IPv4 ACL applied to the Control Plane mgmt VRF:

```
switch# show access-list hitcounts ip My_ipv4_ACL control-plane vrf mgmt
Statistics for ACL My_ipv4_ACL (ipv4):
VRF mgmt* (control-plane):
    Hit Count  Configuration
    - 10 permit udp any 172.16.1.0/255.255.255.0
    - 20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any
    - 30 permit tcp 172.26.1.0/255.255.255.0 any syn ack dscp 10
    8 40 deny any any any count
* access-list statistics are shared among each combination of
  context type (interface, VLAN, VRF) and direction (in, out, control-plane).
  use 'access-list TYPE NAME copy' to create a uniquely-named access-list.
```

show capacities

Syntax

```
show capacities <FEATURE> [vsx-peer]
```

Description

Shows system capacities and their values for all features or a specific feature.

Command context

Manager (#)

Parameters

<FEATURE>

Specifies a feature. For example, aaa.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Capacities are expressed in user-understandable terms. Thus they may not map to a specific hardware or software resource or component. They are not intended to define a feature exhaustively.

Examples

Showing all available capacities for mirroring:

```
switch# show capacities mirroring
```

System Capacities: Filter Mirroring	
Capacities Name	Value
Maximum number of Mirror Sessions configurable in a system	4
Maximum number of enabled Mirror Sessions in a system	4

Showing all available capacities for MSTP:

```
switch# show capacities mstp
```

System Capacities: Filter MSTP	
Capacities Name	Value
Maximum number of mstp instances configurable in a system	64

Showing all available capacities for VLAN count:

```
switch# show capacities vlan-count
```

System Capacities: Filter VLAN Count	
Capacities Name	Value
Maximum number of VLANs supported in the system	4094

show capacities-status

Syntax

```
show capacities-status <FEATURE>  
    [vsx-peer]
```

Description

Shows system capacities status and their values for all features or a specific feature.

Command context

Manager (#)

Parameters

<FEATURE>

Specifies the feature, for example `aaa` for which to display capacities, values, and status. Required.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing the system capacities status for all features:

```
switch# show capacities-status
```

System Capacities Status		
Capacities Status Name	Value	Maximum
Number of active gateway mac addresses in a system	0	16
Number of aspath-lists configured	0	64
Number of community-lists configured	0	64
...		

show object-group

Syntax

```
show object-group [{ip|ipv6} address | port] [<OBJECT-GROUP-NAME>] [commands] [configuration]
```

Description

Shows information about your defined object groups. When `show object-group` is entered without parameters, information for all object groups is shown. The parameters filter the list of object groups for which information is shown.

Command context

Operator (>) or Manager (#)

Parameters

[{ip|ipv6} address | port]

Specifies the object group type, either `address` for an IP address, or `port`.

<OBJECT-GROUP-NAME>

Specifies the object group name.

[commands]

Specifies that the object group definition is to be shown as the commands and parameters used to create it rather than in tabular form.

[configuration]

Specifies that the user-configured object groups be shown as configured. The output of the command with this parameter may not be the same as what is active on the switch due to a misconfigured object group. See *Examples* in this topic.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing configured object groups:

```
switch# show object-group
Type      Name
Sequence L4 Port(s)/IP Address
-----
IPv4      my_address_group
          10 192.168.0.1
          20 192.168.0.3
Port      my_port_group
          10 eq 80
          20 gt 65525
switch#
switch# show object-group commands
object-group ip address my_address_group
    10 192.168.0.1
    20 192.168.0.3
object-group port my_port_group
    10 eq 80
    20 gt 65525
```

Showing a misconfigured object group:

```
switch# show object-group
Type      Name
Sequence L4 Port(s)/IP Address
-----
! object-group ip address My_ip_object_group user configuration does not match
! the active hardware configuration. Run 'object-group ip address NAME reset'
! to reset the object group to match the active hardware configuration.
IPv4      my_address_group
switch#
switch# show object-group configuration
Type      Name
Sequence L4 Port(s)/IP Address
-----
! object-group ip address My_ip_object_group user configuration does not match
! the active hardware configuration. Run 'object-group ip address NAME reset'
! to reset the object group to match the active hardware configuration.
IPv4      my_address_group
          10 192.168.0.1
          20 192.168.0.3
switch#
switch# show object-group commands
! object-group ip address My_ip_object_group user configuration does not match
! the active hardware configuration. Run 'object-group ip address NAME reset'
! to reset the object group to match the active hardware configuration.
switch#
switch# show object-group commands configuration
! object-group ip address My_ip_object_group user configuration does not match
```

```
! the active hardware configuration. Run 'object-group ip address NAME reset'
! to reset the object group to match the active hardware configuration.
object-group ip address my_address_group
  10 192.168.0.1
  20 192.168.0.3
```

Resetting a misconfigured object group:

```
switch(config)# object-group all reset
switch(config)# exit
switch# show object-group
Type      Name
Sequence L4 Port(s)/IP Address
-----
IPv4      my_address_group
switch#
switch# show object-group configuration
Type      Name
Sequence L4 Port(s)/IP Address
-----
IPv4      my_address_group
```

Classifier policies overview

Classifier policies let a network administrator define sets of rules based on network traffic addressing or other header content, and use these rules to restrict or alter the passage of traffic through the switch. Choosing the rule criteria is called Classification, and one such rule, or list, is called a policy. Classification is achieved by creating a traffic class. The three types of classes (MAC, IPv4, and IPv6) are each focused on relevant frame/packet characteristics. Classes can be configured to match or ignore almost any frame or packet header field. Network traffic passing through a switch can be classified based on many different frame/packet characteristics including, but not limited to:

- Frame ingress VLAN ID
- Source and/or destination Ethernet MAC, IPv4, or IPv6 address
- Layer 2 (EtherType) and Layer 3 (IP) protocol
- Layer 4 application ports

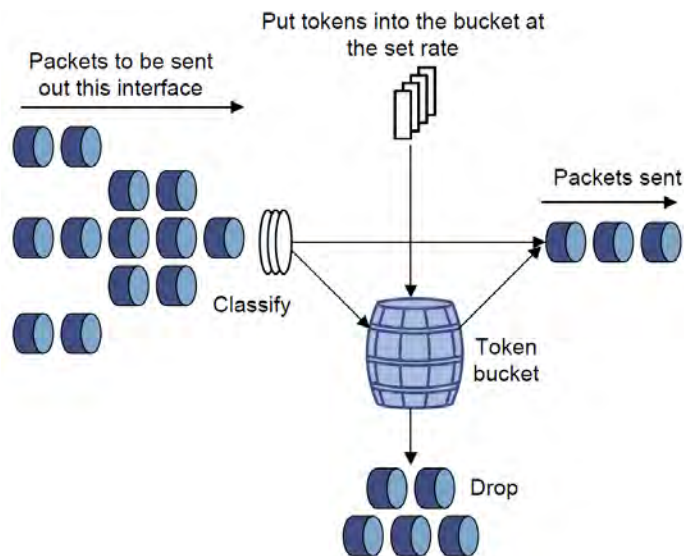
A policy contains one or more policy entries, which are listed according to priority by sequence number. A single policy entry contains a class and corresponding policy action. Policy action is taken on traffic matched by its corresponding class. A policy can be applied to an individual front plane port, a Link Aggregation Group (LAG) interface, or a VLAN.



IMPORTANT: See also [ACL and Policy hardware resource considerations](#).

Traffic policing

Traffic policing supports policing of the inbound traffic. A typical application of traffic policing is to supervise the specification of traffic entering a network and limit it within a reasonable range. Another application is to "discipline" the extra traffic to prevent aggressive use of network resources by an application. For example, you can limit bandwidth for HTTP packets to less than 50% of the total. If the traffic of a session exceeds the limit, traffic policing can drop the packets or reset the IP precedence of the packets. In the following illustrated example, outbound traffic is policed:



Traffic policing is widely used in policing traffic entering the ISP networks. It can classify the policed traffic and take predefined policing actions on each packet depending on the evaluation result:

- Forwarding the packet if the evaluation result is "conforming."
- Dropping the packet if the evaluation result is "excess."
- Forwarding the packet with its precedence remarked when the evaluation result is "conforming."

Types of policy actions

The policy actions are broadly classified in the following categories:

- Remark actions
- Police actions
- Other actions

Each policy entry can have a combination of policy actions from these multiple categories, which are executed in the order of configuration.

Remark actions

This category contains the following actions:

- **Priority Code Point (PCP):** 3-bit field in layer 2 802.1Q header refers to a class of service and maps to a frame priority level.
- **IP precedence:** 3-bit field in IP header which denotes the importance or priority of the datagram.
- **IP Differentiated Services Code Point (DSCP):** 6-bit field in IP header for packet classification.
- **Local Priority:** Change the internal priority used to queue the packets for transmission. Local priority can be used to rewrite the priority of traffic classes local to the system based on the QoS mapping settings without changing the IP header or the 802.1Q header. Remark actions other than local priority only change packets as they leave the switch. The local priority action can be combined with the other remark actions to remark packets and change the internal priority to reflect the new priority.

Police actions

Traffic policing meters inbound traffic on an interface or VLAN based on the following traffic parameters:

- Committed information rate (CIR): Bandwidth limit for guaranteed traffic.
- Committed burst size (CBS): Maximum packet size permitted for bursts of data that exceed the CIR.

Based on these parameters, packets are dropped when traffic exceeds the bandwidth limit (CIR) and the burst size for guaranteed traffic (CBS).

Other actions

Other actions include Drop: Drop the packet, and Mirror: Mirror the packets to a specified mirroring session. For details, see the *Monitoring Guide*.

How policy matching works

A policy can be applied to an interface or VLAN to affect/control traffic arriving on that interface or VLAN (inbound (ingress)).

A single policy entry matches on one or more characteristics of the particular traffic type and has a configured action to continue through the switch.

This matching occurs by beginning with the entry with the lowest sequence number. The entry is then compared against the incoming frame to its particular match characteristics. If there is a match, the action is taken.

If there is no match, the match characteristics of the next sequence are compared to the relevant frame/packet details. If there is a match, the specified actions are taken.

This process continues until a match is found; otherwise, the packet is permitted to flow through the switch unaltered. The "implicit permit" behavior of policy matching differs from the "implicit deny" behavior of ACL matching.

Active class configuration versus user-specified configuration

The output of the `show class` command displays the active class configurations. Active class configurations are the classes that have been configured and accepted by the system.

The output of the `show class` command with the `configuration` parameter, displays the classes that have been configured by the user.

Discrepancies might occur between the active class configurations and the user-specified configurations. In the user-specified class configurations, unsupported command parameters may have been configured, or class can be modified after policy application and may have been unsuccessful due to lack of hardware resources.

To determine if a discrepancy exists between what was configured and what is active, run any variant of the `show class` command. If the active classes and configured classes are not the same, a warning message is displayed.

```
! class MY_CLASS user configuration does not match active configuration.  
! run 'class TYPE NAME reset' to reset class to match active configuration.
```

If the configured class is processing and you entered the `show class` command with parameters, the following in-progress message is displayed:

```
! class ip MY_CLASS user configuration currently being processed
! run 'class TYPE NAME reset' to reset class to match active configuration.
```

If the configured class is processing and you entered the `show class` command without parameters, the following in-progress message is displayed:

```
% Warning: MY_CLASS user configuration currently being processed
% run 'class TYPE NAME reset' to reset class to match active configuration.
```

If the warning message or in-progress message is displayed, additional changes may be made until the error message is no longer displayed. Or you can use the `class {all|ip <class-name>|ipv6 <class-name>|mac <class-name>} reset` command to change the user-specified configuration to match the active configuration.



NOTE: The `show running-config` command also shows a warning about classes that are in progress or failed.

Example

Resetting the user-specified class configuration to the active configuration:

```
switch(config)# class all reset
```

Active policy configuration versus user-specified configuration

The output of the `show policy` command displays the active policy configurations. Active policy configurations are the policies that have been configured and accepted by the system. With applied policies, the active configuration displays the interfaces on which the policies have successfully been programmed in hardware.

The output of the `show policy` command with the `configuration` parameter, displays the policies that have been configured by the user.

Discrepancies might exist between the active policy configurations and the user-specified configurations. In the user-specified policy configurations, unsupported command parameters might have been configured, or an application of a policy might have been unsuccessful because of a lack of hardware resources.

To determine if a discrepancy exists between the configuration and what is active, run any variant of the `show policy` command. If the active policies and configured policies are not the same, a warning message is displayed in the output of the `show` command.

```
! policy MY_POLICY user configuration does not match active configuration.
! run 'policy NAME reset' to reset policy to match active configuration.
```

The switch displays an `in progress` message while it processes the configured policy:

```
! policy MY_POLICY user configuration currently being processed
! run 'policy NAME reset' to reset policy to match active configuration.
```

If the warning message or in progress message is displayed, additional changes may be made until the error message is no longer displayed. Or you can use the `policy <policy-name> reset` command to change the user-specified configuration to match the active configuration.

Example

Resetting MY_POLICY:

```
switch(config)# policy MY_POLICY reset
```

Classifier policy commands

Classifier policy application

Classifier policies can be applied as follows:

Policy type	IPv4	IPv4	IPv6	IPv6	IPv4+6	IPv4+6	MAC	MAC
Direction	In	Out	In	Out	In	Out	In	Out
L2 interface (port)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
L2 LAG	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
VLAN	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes



NOTE: The following match criteria is not supported. If this match criteria is attempted to be configured, an error message will be displayed and the action will not be completed.

PCP on MAC classes

apply policy (Context: config)

Syntax

```
apply policy <POLICY-NAME> in
```

```
no apply policy <POLICY-NAME> in
```

Description

Applies a policy to the global config context.

Only one policy can be globally applied at a time. Applying a policy globally again, replaces the previous globally applied policy.

The `no` form of this command removes application of the global policy.

Command context

```
config
```

Parameters

<POLICY-NAME>

Specifies the policy to apply.

in

Selects the inbound (ingress) traffic direction.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Applying policy global1 to the global config context:


```
switch(config)# apply policy global1 in
```

Removing application of policy global1 from the global config context:

```
switch(config)# no apply policy global1 in
```

apply policy (Contexts: config-if, config-vlan)

Syntax

Context: config-if:

```
apply policy <POLICY-NAME> {in|out}
```

```
no apply policy <POLICY-NAME> {in|out}
```

Context: config-vlan:

```
apply policy <POLICY-NAME> {in|out}
```

```
no apply policy <POLICY-NAME> {in|out}
```

Description

Applies a policy to the current interface or VLAN context.

Only one direction of a policy can be applied to an interface or VLAN at a time, thus using the apply command on an interface or VLAN with an already-applied policy of the same direction will replace the currently applied policy.



NOTE: The VLAN context supports the `in` and `out` directions.

The `no` form of this command removes a policy from the interface or VLAN specified by the current context.

Command context

config-if

config-vlan

Parameters

<POLICY-NAME>

Specifies the policy to apply.

in

Selects the inbound (ingress) traffic direction.

out

Selects the outbound (egress) traffic direction.

Authority

Administrators or local user group members with execution rights for this command.

Usage (applies to `config-vlan` context)

Only one policy may be applied to a VLAN at a time. Therefore, using the `apply policy` command on a VLAN with an already-applied policy of the same type, will replace the applied policy.

Examples

Applying a policy to an interface (ingress):

```
switch(config)# interface 1/1/1
switch(config-if)# apply policy MY_POLICY in
```

Applying a policy to an interface (egress):

```
switch(config)# interface 1/1/2
switch(config-if)# apply policy MY_POLICY2 out
```

Applying a policy to an interface range (egress):

```
switch(config)# interface 1/1/2-1/1/5
switch(config-if-<1/1/2-1/1/5>)# apply policy MY_POLICY3 out
```

Removing a policy from an interface range (egress)

```
switch(config)# 1/1/2-1/1/5
switch(config-if)# no apply policy MY_POLICY3 out
```

Applying a policy to a VLAN (ingress):

```
switch(config)# vlan 10
switch(config-vlan-10)# apply policy MY_POLICY in
```

Applying a policy to multiple VLANs (egress):

```
switch(config)# vlan 20,30
switch(config-vlan-<20,30>)# apply policy MY_POLICY2 out
```

Removing a policy from a VLAN (ingress):

```
switch(config)# vlan 10
switch(config-vlan-10)# no apply policy MY_POLICY in
```

class copy

Syntax

```
class {ip|ipv6|mac} <CLASS-NAME> copy <DESTINATION-CLASS>
```

Description

Copies a IPv4, IPv6, or MAC class to a new destination class or overwrites an existing class. Copying a class copies all entries as well.

Command context

config

Parameters

{ip|ipv6|mac} <CLASS-NAME>

Specifies the type and name of the class to be copied.

<DESTINATION-CLASS>

Specifies the name of the destination class.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Copying an IPv4 class. Copying a class with entries copies all its entries as well:

```
switch(config)# class ip MY_IP_CLASS copy MY_IP_CLASS2
switch(config)# do show class
```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address	Destination L4 Port(s)	Additional Parameters
IPv4	MY_IP_CLASS	11	ignore	any	udp					
		21	match	192.168.0.1	tcp					
				192.168.0.2						
IPv4	MY_IP_CLASS2	11	ignore	any	udp					
		21	match	192.168.0.1	tcp					
				192.168.0.2						

Copying an IPv6 class:

```
switch(config)# class ipv6 MY_IPV6_CLASS copy MY_IPV6_CLASS2
switch(config)# do show class
```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address	Destination L4 Port(s)	Additional Parameters
IPv6	MY_IPV6_CLASS	2	ignore	any	udp					
				any						
IPv6	MY_IPV6_CLASS2	2	ignore	any	udp					
				any						

Copying a MAC class:

```
switch(config)# class mac MY_MAC_CLASS copy MY_MAC_CLASS2
switch(config)# do show class
```

Type	Name	Sequence	Comment
------	------	----------	---------

	Action	EtherType
	Source MAC Address	
	Destination MAC Address	
	Additional Parameters	

MAC	MY_MAC_CLASS	
	2 ignore	arp
	any	
	any	

MAC	MY_MAC_CLASS2	
	2 ignore	arp
	any	
	any	

class ip

Syntax

Syntax to create an IPv4 class and enter its context. Plus syntax to remove a class:

```
class ip <CLASS-NAME>
```

```
no class ip <CLASS-NAME>
```

Syntax (within the class context) for creating or removing class entries for protocols ah, gre, esp, igmp, ospf, pim (ip is available as an alias for any):

```
[<SEQUENCE-NUMBER>]
{match|ignore}
{any|ip|ah|gre|esp|igmp|ospf|pim|<IP-PROTOCOL-NUM>}
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]]}
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]]}
[dscp <DSCP-SPECIFIER>] [ecn <ECN-VALUE>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [ttl <TTL-VALUE>] [count]

no <SEQUENCE-NUMBER>
```

Syntax (within the class context) for creating or removing class entries for protocols sctp, tcp, udp:

```
[<SEQUENCE-NUMBER>]
{match|ignore}
{sctp|tcp|udp}
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]]}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]]}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]
[urg] [ack] [psh] [rst] [syn] [fin] [established]
[dscp <DSCP-SPECIFIER>] [ecn <ECN-VALUE>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [ttl <TTL-VALUE>] [count]

no <SEQUENCE-NUMBER>
```

Syntax (within the class context) for creating or removing class entries for protocol icmp:

```
[<SEQUENCE-NUMBER>]
{match|ignore}
{icmp}
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]]}
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]]}
[icmp-type {echo|echo-reply|<ICMP-TYPE-VALUE>}] [icmp-code <ICMP-CODE-VALUE>]
[dscp <DSCP-SPECIFIER>] [ecn <ECN-VALUE>] [ip-precedence <IP-PRECEDENCE-VALUE>]
```

```
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [ttl <TTL-VALUE>] [count]
```

```
no <SEQUENCE-NUMBER>
```

Syntax (within the class context) for class entry comments:

```
[<SEQUENCE-NUMBER>] comment <TEXT-STRING>
```

```
no <SEQUENCE-NUMBER> comment
```

Description

Creates or modifies an IPv4 traffic class to match specified packets. Class is composed of one or more class entries ordered and prioritized by sequence numbers. With this command, the class can classify traffic based on IPv4 header information.

The `no` form of the command can be used to delete either an IPv4 traffic class (use `no` with the class command) or an individual IPv4 traffic class entry (use `no` with the sequence number).

Command context

```
config
```

The `class ip <CLASS-NAME>` command takes you into the `config-class-ip` context where you enter the class entries.

Parameters

ip

Specifies create or modify an IPv4 class.

<CLASS-NAME>

Specifies the name of this class.

<SEQUENCE-NUMBER>

Specifies a sequence number for the class entry. Optional. Range: 1-4294967295.

{match|ignore}

Creates a rule to match or ignore specified packets.

<IP-PROTOCOL-NUM>

Specifies the protocol as its Internet Protocol number. For example, 2 corresponds to the IGMP protocol. Range: 0 to 255.

{any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}] }

Specifies the source IPv4 address.

- `any` - specifies any source IPv4 address.
- `<SRC-IP-ADDRESS>` - specifies the source IPv4 host address.
 - `<PREFIX-LENGTH>` - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32.
 - `<SUBNET-MASK>` - specifies the address bits to mask (dotted decimal notation).

{any|<DST-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}] }

Specifies the destination IPv4 address.

- `any` - specifies any destination IPv4 address.
- `<DST-IP-ADDRESS>` - specifies the destination IPv4 host address.
 - `<PREFIX-LENGTH>` - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32.
 - `<SUBNET-MASK>` - specifies the address bits to mask (dotted decimal notation).

[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]

Specifies the port or port range. Port numbers are in the range of 0 to 65535.

- `eq <PORT>` - specifies the Layer 4 port.
- `gt <PORT>` - specifies any Layer 4 port greater than the indicated port.
- `lt <PORT>` - specifies any Layer 4 port less than the indicated port.
- `range <MIN-PORT> <MAX-PORT>` - specifies the Layer 4 port range.

urg

Specifies matching on the TCP Flag: Urgent.

ack

Specifies matching on the TCP Flag: Acknowledgment.

psh

Specifies matching on the TCP Flag: Push buffered data to receiving application.

rst

Specifies matching on the TCP Flag: Reset the connection.

syn

Specifies matching on the TCP Flag: Synchronize sequence numbers.

fin

Specifies matching on the TCP Flag: Finish connection.

established

Specifies matching on the TCP Flag: Established connection.

dscp <DSCP-SPECIFIER>

Specifies the Differentiated Services Code Point (DSCP), either a numeric `<DSCP-VALUE>` (0 to 63) or one of these keywords:

- `AF11` - DSCP 10 (Assured Forwarding Class 1, low drop probability)
- `AF12` - DSCP 12 (Assured Forwarding Class 1, medium drop probability)
- `AF13` - DSCP 14 (Assured Forwarding Class 1, high drop probability)
- `AF21` - DSCP 18 (Assured Forwarding Class 2, low drop probability)
- `AF22` - DSCP 20 (Assured Forwarding Class 2, medium drop probability)
- `AF23` - DSCP 22 (Assured Forwarding Class 2, high drop probability)

- AF31 - DSCP 26 (Assured Forwarding Class 3, low drop probability)
- AF32 - DSCP 28 (Assured Forwarding Class 3, medium drop probability)
- AF33 - DSCP 30 (Assured Forwarding Class 3, high drop probability)
- AF41 - DSCP 34 (Assured Forwarding Class 4, low drop probability)
- AF42 - DSCP 36 (Assured Forwarding Class 4, medium drop probability)
- AF43 - DSCP 38 (Assured Forwarding Class 4, high drop probability)
- CS0 - DSCP 0 (Class Selector 0: Default)
- CS1 - DSCP 8 (Class Selector 1: Scavenger)
- CS2 - DSCP 16 (Class Selector 2: OAM)
- CS3 - DSCP 24 (Class Selector 3: Signaling)
- CS4 - DSCP 32 (Class Selector 4: Realtime)
- CS5 - DSCP 40 (Class Selector 5: Broadcast video)
- CS6 - DSCP 48 (Class Selector 6: Network control)
- CS7 - DSCP 56 (Class Selector 7)
- EF - DSCP 46 (Expedited Forwarding)

ecn *<ECN-VALUE>*

Specifies an Explicit Congestion Notification value. Range: 0 to 3.

ip-precedence *<IP-PRECEDENCE-VALUE>*

Specifies an IP precedence value. Range: 0 to 7.

tos *<TOS-VALUE>*

Specifies the Type of Service value. Range: 0 to 31.

fragment

Specifies a fragment packet.

vlan *<VLAN-ID>*

Specifies VLAN tag to match on. 802.1Q VLAN ID.



NOTE: This parameter cannot be used in any class that will be applied to a VLAN.

ttl *<TTL-VALUE>*

Specifies a time-to-live (hop limit) value. Range: 0 to 255.

count

Keeps the hit counts of the number of packets matching this class entry.

[<SEQUENCE-NUMBER>] comment *<TEXT-STRING>*

Adds a comment to a class entry. The **no** form removes only the comment from the class entry.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- Entering an existing `<CLASS-NAME>` value will cause the existing class to be modified, with any new `<SEQUENCE-NUMBER>` value creating an additional class entry, and any existing `<SEQUENCE-NUMBER>` value replacing the existing class entry with the same sequence number.
- If no sequence number is specified, a new class entry will be appended to the end of the class with a sequence number equal to the highest class entry currently in the list plus 10.
- If the `<IP-PROTOCOL-NUM>` parameter is used instead of a protocol name, ensure that any needed class entry-definition parameters specific to the selected protocol are also provided.

Examples

Creating an IPv4 class with three entries:

```
switch(config)# class ip MY_IP_CLASS
switch(config-class-ip)# 10 match icmp any any
switch(config-class-ip)# 20 ignore udp any any
switch(config-class-ip)# 30 match tcp 192.168.0.1 192.168.0.2
switch(config-class-ip)# exit

switch(config)# do show class
Type          Name
Sequence      Comment
              Action                    L3 Protocol
              Source IP Address         Source L4 Port(s)
              Destination IP Address     Destination L4 Port(s)
              Additional Parameters
-----
IPv4          MY_IP_CLASS
10 match                    icmp
   any
   any
20 ignore                    udp
   any
   any
30 match                    tcp
   192.168.0.1
   192.168.0.2
```

Adding a comment to an existing IPv4 class entry:

```
switch(config)# class ip MY_IP_CLASS
switch(config-class-ip)# 30 comment myipClass
switch(config-class-ip)# exit

switch(config)# do show class
Type          Name
Sequence      Comment
              Action                    L3 Protocol
              Source IP Address         Source L4 Port(s)
              Destination IP Address     Destination L4 Port(s)
              Additional Parameters
-----
IPv4          MY_IP_CLASS
10 match                    icmp
   any
   any
20 ignore                    udp
   any
   any
30 myipClass
```



```

match
192.168.0.1
192.168.0.2
tcp

```

Removing a comment from an existing IPv4 class entry:

```

switch(config)# class ip MY_IP_CLASS
switch(config-class-ip)# no 30 comment
switch(config-class-ip)# exit

switch(config)# do show class
Type      Name
Sequence Comment
          Action
          Source IP Address
          Destination IP Address
          Additional Parameters
-----
IPv4      MY_IP_CLASS
10 match
   any
   any
20 ignore
   any
   any
30 match
   192.168.0.1
   192.168.0.2
          icmp
          udp
          tcp

```

Replacing an IPv4 class entry in an existing class:

```

switch(config)# class ip MY_IP_CLASS
switch(config-class-ip)# 10 match igmp any any
switch(config-class-ip)# exit

switch(config)# do show class
Type      Name
Sequence Comment
          Action
          Source IP Address
          Destination IP Address
          Additional Parameters
-----
IPv4      MY_IP_CLASS
10 match
   any
   any
20 ignore
   any
   any
30 match
   192.168.0.1
   192.168.0.2
          igmp
          udp
          tcp

```

Removing an IPv4 class entry:

```

switch(config)# class ip MY_IP_CLASS
switch(config-class-ip)# no 10
switch(config-class-ip)# exit

switch(config)# do show class
Type      Name
Sequence Comment
          Action
          L3 Protocol

```

		Source IP Address Destination IP Address Additional Parameters	Source L4 Port(s) Destination L4 Port(s)
IPv4		MY_IP_CLASS	
	20	ignore any any	udp
	30	match 192.168.0.1 192.168.0.2	tcp

Removing an IPv4 class. Removing a class with entries removes all its entries as well. If a class associated with a policy entry (or multiple policy entries) is removed, the corresponding entries are also removed.



NOTE: The corresponding entries are only removed if the class is unused by all policy entries.

```
switch(config)# no class ip MY_IP_CLASS

switch(config)# do show class
No Class found.
```

class ipv6

Syntax

Syntax to create an IPv6 class and enter its context. Plus syntax to remove a class:

```
class ipv6 <CLASS-NAME>

no class ipv6 <CLASS-NAME>
```

Syntax (within the class context) for creating or removing class entries for protocols ah, gre, esp, igmp, ospf, pim (ipv6 is available as an alias for any):

```
[<SEQUENCE-NUMBER>]
{match|ignore}
{any|ipv6|ah|gre|esp|igmp|ospf|pim|<IP-PROTOCOL-NUM>}
{any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}]}
{any|<DST-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}]}
[dscp <DSCP-SPECIFIER>] [ecn <ECN-VALUE>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [ttl <TTL-VALUE>] [count]

no <SEQUENCE-NUMBER>
```

Syntax (within the class context) for creating or removing class entries for protocols sctp, tcp, udp:

```
[<SEQUENCE-NUMBER>]
{match|ignore}
{sctp|tcp|udp}
{any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}]}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]
{any|<DST-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}]}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]
[urg] [ack] [psh] [rst] [syn] [fin] [established]
[dscp <DSCP-SPECIFIER>] [ecn <ECN-VALUE>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [ttl <TTL-VALUE>] [count]

no <SEQUENCE-NUMBER>
```

Syntax (within the class context) for creating or removing class entries for protocol icmpv6:

```
[<SEQUENCE-NUMBER>]
{permit|deny}
{icmpv6}
{any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>} ] }
{any|<DST-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>} ] }
[icmp-type {echo|echo-reply|<ICMP-TYPE-VALUE>}] [icmp-code <ICMP-CODE-VALUE>]
[dscp <DSCP-SPECIFIER>] [ecn <ECN-VALUE>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [ttl <TTL-VALUE>] [count]

no <SEQUENCE-NUMBER>
```

Syntax (within the class context) for class entry comments:

```
[<SEQUENCE-NUMBER>] comment <TEXT-STRING>

no <SEQUENCE-NUMBER> comment
```

Description

Creates or modifies an IPv6 traffic class to match specified packets. Class is composed of one or more class entries ordered and prioritized by sequence numbers. With this command, each class can classify traffic based on IPv6 header information.

The `no` form of the command deletes either an IPv6 traffic class (use `no` with the `class` command) or an individual IPv6 traffic class entry (use `no` with the sequence number).

Command context

`config`

The `class ipv6 <CLASS-NAME>` command takes you into the `config-class-ipv6` command context where you enter the class entries.

Parameters

`ipv6`

Specifies create or modify an IPv6 class.

`<CLASS-NAME>`

Specifies the name of this class.

`<SEQUENCE-NUMBER>`

Specifies a sequence number for the class entry. Optional. Range: 1-4294967295.

`{match|ignore}`

Creates a rule to match or ignore specified packets.

`<IP-PROTOCOL-NUM>`

Specifies the protocol as its Internet Protocol number. For example, 2 corresponds to the IGMP protocol. Range: 0 to 255.

`{any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>} ] }`

Specifies the source IPv6 address.

- `any` - specifies any source IPv6 address.
- `<SRC-IP-ADDRESS>` - specifies the source IPv4 host address.

- `<PREFIX-LENGTH>` - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32.
- `<SUBNET-MASK>` - specifies the address bits to mask (dotted decimal notation).

{ any | `<DST-IP-ADDRESS>` [/ { `<PREFIX-LENGTH>` | `<SUBNET-MASK>` }] }

Specifies the destination IPv4 address.

- `any` - specifies any destination IPv6 address.
- `<DST-IP-ADDRESS>` - specifies the destination IPv6 host address.
- `<PREFIX-LENGTH>` - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32.
- `<SUBNET-MASK>` - specifies the address bits to mask (dotted decimal notation).

[{ `eq` | `gt` | `lt` } `<PORT>` | `range` `<MIN-PORT>` `<MAX-PORT>`]

Specifies the port or port range. Port numbers are in the range of 0 to 65535.

- `eq` `<PORT>` - specifies the Layer 4 port.
- `gt` `<PORT>` - specifies any Layer 4 port greater than the indicated port.
- `lt` `<PORT>` - specifies any Layer 4 port less than the indicated port.
- `range` `<MIN-PORT>` `<MAX-PORT>` - specifies the Layer 4 port range.

urg, ack, psh, rst, syn, fin, established

These TCP flag matching parameters are not supported.

dscp `<DSCP-SPECIFIER>`

Specifies the Differentiated Services Code Point (DSCP), either a numeric `<DSCP-VALUE>` (0 to 63) or one of these keywords:

- AF11 - DSCP 10 (Assured Forwarding Class 1, low drop probability)
- AF12 - DSCP 12 (Assured Forwarding Class 1, medium drop probability)
- AF13 - DSCP 14 (Assured Forwarding Class 1, high drop probability)
- AF21 - DSCP 18 (Assured Forwarding Class 2, low drop probability)
- AF22 - DSCP 20 (Assured Forwarding Class 2, medium drop probability)
- AF23 - DSCP 22 (Assured Forwarding Class 2, high drop probability)
- AF31 - DSCP 26 (Assured Forwarding Class 3, low drop probability)
- AF32 - DSCP 28 (Assured Forwarding Class 3, medium drop probability)
- AF33 - DSCP 30 (Assured Forwarding Class 3, high drop probability)
- AF41 - DSCP 34 (Assured Forwarding Class 4, low drop probability)
- AF42 - DSCP 36 (Assured Forwarding Class 4, medium drop probability)
- AF43 - DSCP 38 (Assured Forwarding Class 4, high drop probability)

- **CS0** - DSCP 0 (Class Selector 0: Default)
- **CS1** - DSCP 8 (Class Selector 1: Scavenger)
- **CS2** - DSCP 16 (Class Selector 2: OAM)
- **CS3** - DSCP 24 (Class Selector 3: Signaling)
- **CS4** - DSCP 32 (Class Selector 4: Real time)
- **CS5** - DSCP 40 (Class Selector 5: Broadcast video)
- **CS6** - DSCP 48 (Class Selector 6: Network control)
- **CS7** - DSCP 56 (Class Selector 7)
- **EF** - DSCP 46 (Expedited Forwarding)

ecn *<ECN-VALUE>*

Specifies an Explicit Congestion Notification value. Range: 0 to 3.

ip-precedence *<IP-PRECEDENCE-VALUE>*

Specifies an IP precedence value. Range: 0 to 7.

tos *<TOS-VALUE>*

Specifies the Type of Service value. Range: 0 to 31.

fragment

Specifies a fragment packet.

vlan *<VLAN-ID>*

Specifies VLAN tag to match on. 802.1Q VLAN ID.



NOTE: This parameter cannot be used in any class that will be applied to a VLAN.

ttl *<TTL-VALUE>*

Specifies a time-to-live (hop limit) value. Range: 0 to 255.

count

Keeps the hit counts of the number of packets matching this class entry.

[<SEQUENCE-NUMBER>] comment *<TEXT-STRING>*

Adds a comment to a class entry. The **no** form removes only the comment from the class entry.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- If you enter an existing `<CLASS-NAME>` value, the existing class is modified with any new `<SEQUENCE-NUMBER>` value. This action creates an additional class entry. Any existing `<SEQUENCE-NUMBER>` value replaces the existing class entry with the same sequence number.
- If no sequence number is specified, a new class entry is appended to the end of the class with a sequence number equal to the highest class entry currently in the list plus 10.
- If the `<IP-PROTOCOL-NUM>` parameter is used instead of a protocol name, ensure that any needed class entry-definition parameters specific to the selected protocol are also provided.

Examples

Creating an IPv6 class with two entries:

```
switch(config)# class ipv6 MY_IPV6_CLASS
switch(config-class-ipv6)# 10 match icmpv6 any any
switch(config-class-ipv6)# 20 ignore udp any any
switch(config-class-ipv6)# exit

switch(config)# do show class
Type      Name
Sequence  Comment
          Action
          Source IP Address
          Destination IP Address
          Additional Parameters
-----
IPv6      MY_IPV6_CLASS
          10 match
          any
          any
          20 ignore
          any
          any
```

Adding a comment to an existing IPv6 class entry:

```
switch(config)# class ipv6 MY_IPV6_CLASS
switch(config-class-ipv6)# 10 match icmpv6 any any
switch(config-class-ipv6)# 20 ignore udp any any
switch(config-class-ipv6)# 20 comment myipv6class
switch(config-class-ipv6)# exit

switch(config)# do show class
Type      Name
Sequence  Comment
          Action
          Source IP Address
          Destination IP Address
          Additional Parameters
-----
IPv6      MY_IPV6_CLASS
          10 match
          any
          any
          20 myipv6class
          ignore
          any
          any
```

Removing a comment from an existing IPv6 class entry:

```
switch(config)# class ipv6 MY_IPV6_CLASS
switch(config-class-ipv6)# no 20 comment
switch(config-class-ipv6)# exit

switch(config)# do show class
Type          Name
Sequence      Comment
              Action
              Source IP Address
              Destination IP Address
              Additional Parameters
-----
IPv6          MY_IPV6_CLASS
10 match
   any
   any
20 ignore
   any
   any
```

Type	Name	Sequence	Comment	Action	Source IP Address	Destination IP Address	Additional Parameters	L3 Protocol	Source L4 Port(s)	Destination L4 Port(s)
IPv6	MY_IPV6_CLASS	10	match					icmpv6		
			any							
			any							
		20	ignore					udp		
			any							
			any							

Replacing an IPv6 class entry in an existing IPv6 class:

```
switch(config)# class ipv6 MY_IPV6_CLASS
switch(config-class-ipv6)# 10 match any any 1020::
switch(config-class-ipv6)# exit

switch(config)# do show class
Type          Name
Sequence      Comment
              Action
              Source IP Address
              Destination IP Address
              Additional Parameters
-----
IPv6          MY_IPV6_CLASS
10 match
   any
   1020::
20 ignore
   any
   any
```

Type	Name	Sequence	Comment	Action	Source IP Address	Destination IP Address	Additional Parameters	L3 Protocol	Source L4 Port(s)	Destination L4 Port(s)
IPv6	MY_IPV6_CLASS	10	match					any		
			any							
			1020::							
		20	ignore					udp		
			any							
			any							

Removing an IPv6 class entry:

```
switch(config)# class ipv6 MY_IPV6_CLASS
switch(config-class-ipv6)# no 10
switch(config-class-ipv6)# exit

switch(config)# do show class
Type          Name
Sequence      Comment
              Action
              Source IP Address
              Destination IP Address
              Additional Parameters
-----
IPv6          MY_IPV6_CLASS
20 ignore
   any
   any
```

Type	Name	Sequence	Comment	Action	Source IP Address	Destination IP Address	Additional Parameters	L3 Protocol	Source L4 Port(s)	Destination L4 Port(s)
IPv6	MY_IPV6_CLASS	20	ignore					udp		
			any							
			any							

Removing an IPv6 class. Removing a class with entries removes all its entries as well. If a class associated with a policy entry (or multiple policy entries) is removed, the corresponding entries are also removed.



NOTE: The corresponding entries are only removed if the class is unused by all policy entries.

```
switch(config)# no class ipv6 MY_IPV6_CLASS

switch(config)# do show class
No Class found.
```

class mac

Syntax

```
class mac <CLASS-NAME>

    [<SEQUENCE-NUMBER>]
    {match|ignore}
    {any|<SRC-MAC-ADDRESS>[/<ETHERNET-MASK>]}
    {any|<DST-MAC-ADDRESS>[/<ETHERNET-MASK>]}
    {any|aarp|appletalk|arp|fcoe|fcoe-init|ip|ipv6|ipx-arp|ipx-non-arp|is-is|
        lldp|mpls-multicast|mpls-unicast|q-in-q|rbridge|trill|wake-on-lan|
        <NUMERIC-ETHERTYPE>}
    [pcp <PCP-VALUE>] [vlan <VLAN-ID>] [count]

    [<SEQUENCE-NUMBER>] comment <TEXT-STRING>
```

Description

Creates or modifies a MAC traffic class to match specified packets. Class is composed of one or more class entries ordered and prioritized by sequence numbers. With this command, each class can classify traffic based on MAC header information.

The `no` form of the command can be used to delete either a MAC traffic class (use `no` with the class command) or an individual MAC traffic class entry (use `no` with the sequence number).

Command context

config

The `class mac <CLASS-NAME>` command takes you into the `config-class-mac` context where you enter the class entries.

Parameters

mac

Specifies create or modify a MAC class.

<CLASS-NAME>

Specifies the name of this class.

<SEQUENCE-NUMBER>

Specifies a sequence number for the class entry. Optional. Range: 1-4294967295.

{match|ignore}

Creates a rule to match or ignore specified packets.

comment

Stores the remaining entered text as a class comment.

{ any | <SRC-MAC-ADDRESS> [/ <ETHERNET-MASK>] }

Specifies the source host MAC address (xxxx.xxxx.xxxx), OUI, or the keyword `any`. You can optionally include the following:

<ETHERNET-MASK> - The address bits to mask (xxxx.xxxx.xxxx).

{ any | <DST-MAC-ADDRESS> [/ <ETHERNET-MASK>] }

Specifies the destination host MAC address (xxxx.xxxx.xxxx), OUI, or the keyword `any`. You can optionally include the following:

<ETHERNET-MASK> - The address bits to mask (xxxx.xxxx.xxxx).

Protocol

Select an ethertype protocol from the following (enter one only):

- `any` - Any ethertype protocol
- <NUMERIC-ETHERTYPE> - Enter an EtherType protocol number. Range: 0x600-0xffff.
- Or enter an EtherType protocol name from the following list:
 - `aarp`
 - `appletalk`
 - `arp`
 - `fcoe`
 - `fcoe-init`
 - `ip`
 - `ipv6`
 - `ipx-arpa`
 - `ipx-non-arpa`
 - `is-is`
 - `lldp`
 - `mpls-multicast`
 - `mpls-unicast`
 - `q-in-q`
 - `rbridge`
 - `trill`
 - `wake-on-lan`

pcp <PCP-VALUE>

Not supported.

vlan <VLAN-ID>

Specifies matching on a VLAN ID. Enter a VLAN ID or the VLAN name, if configured.



NOTE: This parameter cannot be used in any class that will be applied to a VLAN.

count

Keeps the hit counts of the number of packets matching this class entry.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating a MAC class:

```
switch(config)# class mac MY_MAC_CLASS
switch(config-class-mac)# match any any lldp
switch(config-class-mac)# ignore any any arp
switch(config-class-mac)# exit
switch(config)# do show class
```

Type	Name	Sequence	Comment	Action	EtherType
				Source MAC Address	
				Destination MAC Address	
				Additional Parameters	
MAC	MY_MAC_CLASS				
		10	match		lldp
			any		
			any		
		20	ignore		arp
			any		
			any		

Adding a comment to an existing MAC class entry:

```
switch(config)# class mac MY_MAC_CLASS
switch(config-class-mac)# 10 comment MY_CLASS_ENTRY
switch(config-class-mac)# exit
switch(config)# do show class
```

Type	Name	Sequence	Comment	Action	EtherType
				Source MAC Address	
				Destination MAC Address	
				Additional Parameters	
MAC	MY_MAC_CLASS				
		10	MY_CLASS_ENTRY		
			match		lldp
			any		
			any		
		20	ignore		arp
			any		
			any		

Removing a comment from an existing MAC class entry:

```
switch(config)# class mac MY_MAC_CLASS
switch(config-class-mac)# no 10 comment MY_CLASS_ENTRY
switch(config-class-mac)# exit
switch(config)# do show class
```

Type	Name	Sequence	Comment	Action	EtherType
				Source MAC Address	
				Destination MAC Address	
				Additional Parameters	

MAC	MY_MAC_CLASS				
		10	match		lldp
			any		
			any		
		20	ignore		arp
			any		
			any		

Replacing a MAC class entry in an existing MAC class:

```
switch(config)# class mac MY_MAC_CLASS
switch(config-class-mac)# 10 match any any any
switch(config-class-mac)# exit
switch(config)# do show class
```

Type	Name	Sequence	Comment	Action	EtherType
				Source MAC Address	
				Destination MAC Address	
				Additional Parameters	

MAC	MY_MAC_CLASS				
		10	match		any
			any		
			any		
		20	ignore		arp
			any		
			any		

Removing a MAC class entry:

```
switch(config)# class mac MY_MAC_CLASS
switch(config-class-mac)# no 1
switch(config-class-mac)# exit
switch(config)# do show class
```

Type	Name	Sequence	Comment	Action	EtherType
				Source MAC Address	
				Destination MAC Address	
				Additional Parameters	

MAC	MY_MAC_CLASS				
		2	ignore		arp
			any		
			any		

Removing a MAC class. Removing a class with entries removes all its entries as well. If a class associated with a policy entry (or multiple policy entries) is removed, the corresponding entries are also removed.



NOTE: The corresponding entries are only removed if the class is unused by all policy entries.

```
switch(config)# no class mac MY_MAC_CLASS
switch(config)# do show class
No Class found.
```

class resequence

Syntax

```
class {ip|ipv6|mac} <CLASS-NAME> resequence <STARTING-SEQUENCE-NUMBER> <INCREMENT>
```

Description

Resequencing numbering in an IPv4, IPv6, or MAC class.

Command context

config

Parameters

{ip|ipv6|mac} <CLASS-NAME>

Specifies the class where you want to resequence class entries.

<STARTING-SEQUENCE-NUMBER>

Specifies the sequence number to start resequencing from.

<INCREMENT>

Specifies how much to increment the sequence numbers by.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Resequencing an IPv4 class:

```
switch(config)# class ip MY_IP_CLASS resequence 1 10
switch(config)# do show class
```

Type	Name	Sequence	Comment	Action	L3 Protocol
				Source IP Address	Source L4 Port(s)
				Destination IP Address	Destination L4 Port(s)
				Additional Parameters	
IPv4	MY_IP_CLASS				
		1	match		igmp
			any		
			any		
		11	ignore		udp
			any		
			any		
		21	match		tcp
			192.168.0.1		
			192.168.0.2		

Resequencing an IPv6 class:

```

switch(config)# class ipv6 MY_IPV6_CLASS resequence 1 1
switch(config-class-ipv6)# exit
switch(config)# do show class

```

Type	Name		
Sequence	Comment	Action	L3 Protocol
		Source IP Address	Source L4 Port(s)
		Destination IP Address	Destination L4 Port(s)
		Additional Parameters	

IPv6	MY_IPV6_CLASS		
1	match		any
	any		
	1020::		
2	ignore		udp
	any		
	any		

Resequencing a MAC class:

```

switch(config)# class mac MY_MAC_CLASS resequence 1 1
switch(config)# do show class

```

Type	Name		
Sequence	Comment	Action	EtherType
		Source MAC Address	
		Destination MAC Address	
		Additional Parameters	

MAC	MY_MAC_CLASS		
1	match		any
	any		
	any		
2	ignore		arp
	any		
	any		

class reset

Syntax

```
class { all | ip <CLASS-NAME> | ipv6 <CLASS-NAME> | mac <CLASS-NAME> } reset
```

Description

Changes the user-specified class configuration to match the active class configuration. Use this command when there is a discrepancy between what the user configured and what is active and accepted by the system.

Command context

config

Parameters

```
{ all | ip <CLASS-NAME> | ipv6 <CLASS-NAME> | mac <CLASS-NAME> }
```

Specifies either `all` classes be reset or specifies the type (`ip` for IPv4, `ipv6` for IPv6 or `mac` for MAC ACL) and name of the class to be reset.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Resetting the user-specified configuration to the active configuration:

```
switch(config)# class all reset
```

clear policy hitcounts

Syntax

```
clear policy hitcounts { all | [<POLICY-NAME>] [[interface <IF-NAME> [in|out]] |  
[vlan <VLAN-ID> [in|out]]] | global }
```

Description

Clears the policy hit count statistics.

Command context

Operator (>) or Manager (#)

Parameters

all

Selects all policies.

<POLICY-NAME>

Specifies the policy name.

interface <IF-NAME>

Specifies the interface name.

vlan <VLAN-ID>

Specifies the VLAN.

in

Specifies the inbound (ingress) traffic direction.

out

Selects the outbound (egress) traffic direction.

global

Selects the globally applied policy.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Clearing policy hit counts and then showing the policy hit counts (statistics):

```
switch# clear policy hitcounts my_policy int 1/1/1 in  
switch# show policy hitcounts my_policy  
Statistics for Policy my_policy:  
Interface 1/1/1* (in):  
Hit Count Configuration  
10 class ipv6 my_class1 action dscp af21 action drop  
0 10 match any any any count
```

```
* policy statistics are shared among each context type (interface, VLAN).
Use 'policy NAME copy' to create a new policy for separate statistics.
```

Clearing the globally applied policy hit counts and then showing the global policy hit counts (statistics):

```
switch# clear policy hitcounts global
switch# show policy hitcounts global
Statistics for Policy my_policy:
Global Policy:
    Hit Count  Configuration
10 class ipv6 my_class1 action mirror
    0 10 match any any any count
* policy statistics are shared among each context type (interface, VLAN).
Use 'policy NAME copy' to create a new policy for separate statistics.
```

Clearing hit counts for policy MY_IPv6_Policy applied to VLAN 10 (ingress):

```
switch# clear policy hitcounts My_IPv6_Policy vlan 10 in
```

Clearing hit counts for all policies:

```
switch# clear policy hitcounts all
```

policy

Syntax

```
policy <POLICY-NAME>
```

```
[<SEQUENCE-NUMBER>]
class {ip|ipv6|mac} <CLASS-NAME>
    action {<REMARK-ACTIONS> | <POLICE-ACTIONS> | <OTHER-ACTIONS>}
    [{<REMARK-ACTIONS> | <POLICE-ACTIONS> | <OTHER-ACTIONS>}]

[<SEQUENCE-NUMBER>]
comment ...
```

Description

Creates or modifies classifier policy and policy entries. A policy is made up of one or more policy entries ordered and prioritized by sequence numbers. Each entry has an IPv4/IPv6/MAC class and zero or more policy actions associated with it.

A policy must be applied using the `apply` command.

The `no` form of the command can be used to delete either a policy (use `no` with the `policy` command) or an individual policy entry (use `no` with the sequence number).

Command context

```
config
```

The `policy` command takes you into the `config-policy` context where you enter the policy entries.

Parameters

<POLICY-NAME>

Specifies the name of the policy.

<SEQUENCE-NUMBER>

Specifies a sequence number for the policy entry. Optional. Range: 1 to 4294967295.

comment

Stores the remaining entered text as a policy entry comment.

class {ip|ipv6|mac} <CLASS-NAME>

Specifies a type of class, `ip` for IPv4, `ipv6` for IPv6 and `mac` for a MAC policy. And specifies a class name.

<REMARK-ACTIONS>

Remark actions can be any of the following options: {`pcp <PRIORITY>` | `ip-precedence <IP-PRECEDENCE-VALUE>` | `dscp <DSCP-VALUE>` | `local-priority <LOCAL-PRIORITY-VALUE>`} where:

pcp <PCP-VALUE>

Specifies Priority Code Point (PCP) value. Range: 0 to 7.

ip-precedence <IP-PRECEDENCE-VALUE>

Specifies the numeric IP precedence value. Range: 0 to 7.

dscp <DSCP-VALUE>

Specifies a Differentiated Services Code Point (DSCP) value. Enter either a numeric value (0 to 63) or a keyword as follows:

- AF11 - DSCP 10 (Assured Forwarding Class 1, low drop probability)
- AF12 - DSCP 12 (Assured Forwarding Class 1, medium drop probability)
- AF13 - DSCP 14 (Assured Forwarding Class 1, high drop probability)
- AF21 - DSCP 18 (Assured Forwarding Class 2, low drop probability)
- AF22 - DSCP 20 (Assured Forwarding Class 2, medium drop probability)
- AF23 - DSCP 22 (Assured Forwarding Class 2, high drop probability)
- AF31 - DSCP 26 (Assured Forwarding Class 3, low drop probability)
- AF32 - DSCP 28 (Assured Forwarding Class 3, medium drop probability)
- AF33 - DSCP 30 (Assured Forwarding Class 3, high drop probability)
- AF41 - DSCP 34 (Assured Forwarding Class 4, low drop probability)
- AF42 - DSCP 36 (Assured Forwarding Class 4, medium drop probability)
- AF43 - DSCP 38 (Assured Forwarding Class 4, high drop probability)
- CS0 - DSCP 0 (Class Selector 0: Default)
- CS1 - DSCP 8 (Class Selector 1: Scavenger)
- CS2 - DSCP 16 (Class Selector 2: OAM)
- CS3 - DSCP 24 (Class Selector 3: Signaling)
- CS4 - DSCP 32 (Class Selector 4: Real time)
- CS5 - DSCP 40 (Class Selector 5: Broadcast video)
- CS6 - DSCP 48 (Class Selector 6: Network control)

- CS7 - DSCP 56 (Class Selector 7)
- EF - DSCP 46 (Expedited Forwarding)

local-priority **<LOCAL-PRIORITY-VALUE>**

Specifies a local priority value. Range: 0 to 7.

<POLICE-ACTIONS>

Police actions can be the following {cir **<RATE-BPS>** cbs **<BYTES>** exceed} where:

cir kbps **<RATE-KBPS>**

Specifies a Committed Information Rate value in Kilobits per second. Range: 1 to 4294967295.

cbs **<BYTES>**

Specifies a Committed Burst Size value in bytes. Range: 1 to 4294967295.

exceed

Specifies action to take on packets that exceed the rate limit.

<OTHER-ACTIONS>

Other actions can be the following:

drop

Specifies drop traffic.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- An applied policy will process a packet sequentially against policy entries in the list until the last policy entry in the list has been evaluated or the packet matches an entry.
- Entering an existing **<POLICY-NAME>** value will cause the existing policy to be modified, with any new **<SEQUENCE-NUMBER>** value creating an additional policy entry, and any existing **<SEQUENCE-NUMBER>** value replacing the existing policy entry with the same sequence number.
- If no sequence number is specified, a new policy entry will be appended to the end of the entry list with a sequence number equal to the highest policy entry currently in the list plus 10.

Examples

Creating a policy with several entries:

```
switch(config)# policy MY_POLICY
switch(config-policy)# 10 class ipv6 MY_CLASS1 action dscp af21 action drop
switch(config-policy)# 20 class ip MY_CLASS3 action mirror 1
switch(config-policy)# exit
switch(config)# do show policy
      Name
Sequence Comment
      Class Type
              action
-----
      MY_POLICY
10      MY_CLASS1 ipv6
              drop
```

```

                dscp AF21

20
    MY_CLASS3 ipv4
        mirror 1

```

Adding a comment to an existing policy entry:

```

switch(config)# policy MY_POLICY
switch(config-policy)# 20 comment MY_TEST_POLICY
switch(config-policy)# exit
switch(config)# do show policy
    Name
Sequence Comment
    Class Type
                action
-----
    MY_POLICY
10
    MY_CLASS1 ipv6
                drop
                dscp AF21

20 MY_TEST_POLICY
    MY_CLASS3 ipv4
        mirror 1

```

Removing a comment from an existing policy entry:

```

switch(config)# policy MY_POLICY
switch(config-policy)# no 20 comment
switch(config-policy)# exit
switch(config)# do show policy
    Name
Sequence Comment
    Class Type
                action
-----
    MY_POLICY
10
    MY_CLASS1 ipv6
                drop
                dscp AF21

20
    MY_CLASS3 ipv4
        mirror 1

```

Adding/Replacing a policy entry in an existing policy:

```

switch(config)# policy MY_POLICY
switch(config-policy)# 10 class ip MY_CLASS3 action drop action dscp af21
switch(config-policy)# exit
switch(config)# do show policy
    Name
Sequence Comment
    Class Type
                action
-----
    MY_POLICY
10
    MY_CLASS3 ipv4
                drop
                dscp AF21

```

```

20
    MY_CLASS3 ipv4
        mirror 1

```

Removing a policy entry:

```

switch(config)# policy MY_POLICY
switch(config-policy)# no 10
switch(config-policy)# exit
switch(config)# do show policy
    Name
    Sequence Comment
    Class Type
        action

```

```

    MY_POLICY
20
    MY_CLASS3 ipv4
        mirror 1

```

Removing a policy:

```

switch(config)# no policy MY_POLICY
switch(config)# do show policy
    Name
    Sequence Comment
    Class Type
        action

```

```

    MY_POLICY2
2
    MY_CLASS3 ipv4
        mirror 1

```

policy copy

Syntax

```
policy <POLICY-NAME> copy <DESTINATION-POLICY>
```

Description

Copies a policy to a new destination policy or overwrites an existing policy. Copying a policy copies all its entries as well.

Command context

config

Parameters

<POLICY-NAME>

Specifies the policy to be copied.

<DESTINATION-POLICY>

Specifies the name of the destination policy.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Copying a policy:

```
switch(config)# policy MY_POLICY copy MY_POLICY2
switch(config)# do show policy
      Name
Sequence Comment
      Class Type
              action
-----
      MY_POLICY
2      my_class3 ipv4
              mirror 1
-----
      MY_POLICY2
2      my_class3 ipv4
              mirror 1
```

policy resequence

Syntax

```
policy <POLICY-NAME> resequence <STARTING-SEQ-NUM> <INCREMENT>
```

Description

Resequences numbering in a policy.

Command context

config

Parameters

<POLICY-NAME>

Specifies the policy where you want to resequence policy entries.

<STARTING-SEQ-NUM>

Specifies the sequence number to start resequencing from.

<INCREMENT>

Specifies how much to increment the sequence numbers by.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Resequencing a policy:

```
switch(config)# policy MY_POLICY resequence 1 1
switch(config)# do show policy
      Name
Sequence Comment
      Class Type
              action
-----
      MY_POLICY
```

```

1      MY_CLASS3 ipv4
        drop
        dscp AF21

2      MY_CLASS3 ipv4
        mirror 1

```

policy reset

Syntax

```
policy <POLICY-NAME> reset
```

Description

Changes the user-specified policy configuration to match the active policy configuration. Use this command when a discrepancy exists between what the user configured and what is active and accepted by the system.

Command context

```
config
```

Parameters

<POLICY-NAME>

Specifies the policy to be reset.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Resetting a policy:

```
switch(config)# policy MY_POLICY reset
```

show class

Syntax

```
show class [ip | ipv6 | mac] [<CLASS-NAME>] [commands] [configuration] [vsx-peer]
```

Description

Shows class configuration information.

Command context

Operator (>) or Manager (#)

Parameters

All parameters are optional.

[ip | ipv6 | mac]

Selects the class type for the display: `ip` for IPv4, `ipv6` for IPv6, or `mac` for MAC classes.

<CLASS-NAME>

Specifies the class name.

commands

Specifies whether to display output as the CLI commands showing the configured class entries.

configuration

Specifies whether to display classes that have been configured by the user, even if they are not active due to issues with the command parameters or hardware issues. This parameter is useful during a mismatch between the entered configuration and the previous successfully programmed (active) classes.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing all class configuration:

```
switch# show class
Type Name
Sequence Comment
      action                      L3 Protocol
Source IP address                Source L4 Port(s)
Destination IP address          Destination L4 Port(s)
Additional Parameters
-----
ipv4 MY_IPV4_CLASS
  10 my first class entry comment
    match                      icmp
    192.168.0.1/255.255.255.0
    192.168.1.1/255.255.255.0
    VLAN: 1
  20 my second class entry comment
    ignore                      tcp
    10.100.0.10/255.255.255.0 < 3000
    10.100.1.10/255.255.255.0 > 2000
    VLAN: 1
-----
```

Showing class configuration for the IPv4 class MY_IPV4_CLASS as CLI commands:

```
switch# show class ip MY_IPV4_CLASS commands
class ip "MY_IPV4_CLASS"
  10 match icmp 192.168.0.1/255.255.255.0 192.168.1.1/255.255.255.0 vlan 1
  10 comment my first class entry comment
  20 ignore tcp 10.100.0.10/255.255.255.0 lt 3000 10.100.1.10/255.255.255.0 gt 2000 vlan 1
  20 comment my second class entry comment
```

show policy

Syntax

Syntax that shows information for all policies:

```
show policy [commands] [configuration] [vsx-peer]
```

Syntax that filters by policies applied to an interface or VLAN:

```
show policy [interface <IF-NAME> [in | out] | vlan <VLAN-ID> [in | out]]  
[commands] [configuration] [vsx-peer]
```

Syntax that filters by the named policy:

```
show policy <POLICY-NAME> [commands] [configuration] [vsx-peer]
```

Syntax that filters by the globally applied policy:

```
show policy global [commands] [configuration] [vsx-peer]
```

Syntax that shows statistical information in the form of hit counts:

```
show policy hitcounts <POLICY-NAME> [interface <IF-NAME> [in | out] |  
vlan <VLAN-ID> [in | out]] [vsx-peer]
```

Syntax that shows statistical information in the form of hit counts for the globally applied policy:

```
show policy hitcounts global [vsx-peer]
```

Description

Shows information about your defined policies and where they have been applied. When `show policy` is entered without parameters, information for all policies is shown. The parameters filter the list of policies for which information is shown.

Available filtering includes:

- The content of a specific policy.
- All policies applied to a specific interface.
- All policies applied to a specific VLAN.
- The globally applied policy.
- The inbound (ingress) or outbound (egress) direction.

To display policy statistics, use the `show policy hitcounts` form of this command.

Command context

Operator (>) or Manager (#)

Parameters

interface <IF-NAME>

Specifies the interface name.

vlan <VLAN-ID>

Specifies the VLAN.

in

Selects the inbound (ingress) traffic direction.

out

Selects the outbound (egress) traffic direction.

<POLICY-NAME>

Specifies the policy name.

commands

Causes the policy definition to be shown as the commands and parameters used to create it rather than in tabular form.

configuration

Causes the user-configured policies be shown as entered, even if the policies are not active due to policy-definition command issues or hardware issues. This parameter is useful if there is a mismatch between the entered configuration and the previous successfully programmed (active) policies configuration.

global

Selects the globally applied policy.

hitcounts

Selects the policy hit counts (statistics).

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing information for all policies:

```
switch# show policy
      Name
Sequence Comment
      Class Type
              action
-----
      my_policy
10 This is my QOS class.
   class1 ipv4
           dscp af21
           drop
-----
```

Showing a policy as commands:

```
switch# show policy commands
policy my_policy
      10 class ip my_class1 action dscp af21 action drop
```

Showing the globally applied policy:

```
switch# show policy global commands
policy my_policy
```



```
    10 class ip my_class1 action drop
apply policy my_policy in
```

Showing policy hit counts (statistics):

```
switch# show policy hitcounts my_policy
Statistics for Policy my_policy:
Interface 1/1/1* (in):
    Hit Count  Configuration
10 class ip my_class1 action dscp af21 action drop
    20  10 match any any any count
* policy statistics are shared among each context type (interface, VLAN).
  Use 'policy NAME copy' to create a new policy for separate statistics.
```

Showing policy hit counts (statistics) for the globally applied policy:

```
switch# show policy hitcounts global
Statistics for Policy my_policy:
Global Policy:
    Hit Count  Configuration
10 class ip my_class1 action mirror
    20  10 match any any any count
* policy statistics are shared among each context type (interface, vlan, VRF).
  use 'policy NAME copy' to create a uniquely-named policy
```

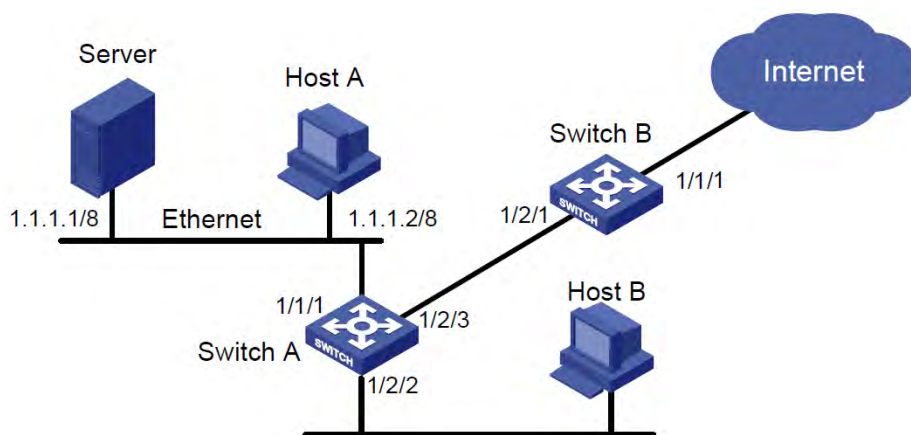
Showing policy hit counts with conform rate from my_policy applied inbound:

```
switch# show policy hitcounts my_policy
Statistics for Policy my_policy:
Interface 1/1/1* (in):
    Hit Count  Configuration
10 class ipv6 my_class1 action cir kbps 1024 cbs 2000 exceed drop [1024 kbps conform ]
    -  10 match any any any
* policy statistics are shared among each context type (interface, VLAN).
  Use 'policy NAME copy' to create a new policy for separate statistics.
```

Intent of the classifier policies configuration example

This example configures traffic policing on:

- A 10-Gbit Ethernet of Switch A meeting the following requirements:
 - Police the rate of packets from the server to 102,400 kbps. Traffic 102,400 kbps or less is forwarded. The traffic more than 102,400 kbps is dropped.
 - Police the rate of packets from Host A to 25,600 kbps. Traffic 25,600 kbps or less is forwarded. The traffic more than 25,600 kbps is dropped.
- A 10-Gbit Ethernet 1/2/1 of Switch B limiting the incoming traffic rate of HTTP packets on 10-Gbit Ethernet 1/1/1 to the data rate of 204,800 kbps and dropping excess packets.



Configuring the classifier policies example

These steps are part of the classifier policies configuration example.

Procedure

1. Configure Switch A.

Create traffic classes named SERVER_TRAFFIC and HOST_A_TRAFFIC for matching the packets from the server and Host A:

```

switch# configure
switch(config)# class ip SERVER_TRAFFIC
switch(config-class-ip)# match any 1.1.1.1 any
switch(config-class-ip)# exit
switch(config)# class ip HOST_A_TRAFFIC

```

```
switch(config-class-ip) # match any 1.1.1.2 any  
switch(config-class-ip) # exit
```

2. Create a classifier policy named RATE_LIMIT_POLICY:

```
switch(config) # policy RATE_LIMIT_POLICY
```

3. Configure the policy RATE_LIMIT_POLICY, so that 102,400 kbps of traffic, matching the class SERVER_TRAFFIC, is forwarded and the excess is dropped:

```
switch(config-policy) # class ip SERVER_TRAFFIC action cir kbps 102400 cbs 0 exceed drop
```

4. Configure the policy RATE_LIMIT_POLICY so that 25,600 kbps of traffic, matching the class HOST_A_TRAFFIC, is forwarded and the excess is dropped:

```
switch(config-policy) # class ip HOST_A_TRAFFIC action cir kbps 25600 cbs 0 exceed drop  
switch(config-policy) # exit
```

5. Apply RATE_LIMIT_POLICY to interface 1/1/1 for the inbound traffic:

```
switch(config) # int 1/1/1  
switch(config-if) # apply policy RATE_LIMIT_POLICY in  
switch(config-if) # exit
```

6. To view the configuration with the RATE_LIMIT_POLICY applied:

```
switch# show running-config  
Current configuration:  
!  
...  
class ip SERVER_TRAFFIC  
  10 match any 1.1.1.1 any  
class ip HOST_A_TRAFFIC  
  10 match any 1.1.1.2 any  
policy RATE_LIMIT_POLICY  
  10 class ip SERVER_TRAFFIC action cir kbps 102400 cbs 0 exceed drop  
  20 class ip HOST_A_TRAFFIC action cir kbps 25600 cbs 0 exceed drop  
interface 1/1/1  
  apply policy RATE_LIMIT_POLICY in
```

7. Configure Switch B.

Create a traffic class named HTTP_TRAFFIC and configure it to match traffic to port 80:

```
switch(config) # class ip HTTP_TRAFFIC  
switch(config-class-ip) # match tcp any any eq 80  
switch(config-class-ip) # exit
```

8. Create a classifier policy named RATE_LIMIT_HTTP:

```
switch(config) # policy RATE_LIMIT_HTTP
```

9. Configure the policy RATE_LIMIT_HTTP so that 204,800 kbps of traffic, matching the class HTTP_TRAFFIC, is forwarded and the excess is dropped:

```
switch(config-policy) # class ip HTTP_TRAFFIC action cir kbps 204800 cbs 0 exceed drop  
switch(config-policy) # exit
```

10. Apply RATE_LIMIT_HTTP to interface 1/1/1 for inbound traffic:

```
switch(config)# int 1/1/1
switch(config-if)# apply policy RATE_LIMIT_HTTP in
switch(config-if)# exit
```

11. Show the running configuration with RATE_LIMIT_HTTP applied:

```
switch# show running-config
Current configuration:
!
...
class ip HTTP_TRAFFIC
  10 match tcp any any eq 80
policy RATE_LIMIT_HTTP
  10 class ip HTTP_TRAFFIC action cir kbps 204800 cbs 0 exceed drop
interface 1/1/1
  apply policy RATE_LIMIT_HTTP in
```

Switches have finite (TCAM and other) hardware resources used in the application of ACLs and classifier policies (including port access policies) to packets being processed in switch hardware. ADC (analytics data collection) also consumes TCAM lookups. Take the considerations described in this chapter into account when deciding what ACL and classifier policy-related features to use at the same time.

TCAM lookups

TCAM lookups are a finite hardware resource used in the application of ACLs and policies (including port access policies) to packets being processed in switch hardware. ADC (analytics data collection) also consumes TCAM lookups. There are a limited number of ACL and policy features that can be enabled at the same time.



NOTE: In the following TCAM lookup lists, "IP" means both IPv4 and IPv6.

There are four TCAM lookups available to use for these features. Each of these features uses one TCAM lookup when enabled. At most, four of these features can be enabled at the same time.

```
Ingress Port IP ACL
Ingress Port MAC ACL
Ingress Port Policy
Ingress Port Access Client Policy
Ingress VLAN IP ACL
Ingress Routed VLAN IP ACL
Ingress VLAN MAC ACL
Ingress VLAN Policy
Ingress Global Policy
Ingress IP ADC
Port Access Client Policy
```

This feature is not classifier related but uses one lookup each from the above group of features:

```
Ingress L2 Tunnel
```

There are four TCAM lookups available for these features. Each of these features uses one TCAM lookup when enabled. At most, four of these features can be enabled at the same time.

```
Egress Port IP ACL
Egress Port MAC ACL
Egress Port Policy
Egress VLAN IP ACL
Egress Routed VLAN IP ACL
Egress VLAN MAC ACL
Egress VLAN Policy
```

Matching precedence order

When a packet is matched by multiple TCAM Lookups with the same action, a precedence order is followed.

For example, if a packet matches an IPv6 ACL with a count action and a MAC ACL with a count action, the IPv6 count action takes precedence and the MAC ACL will not count the packet. However, if a packet matches both an ACL and a policy with count actions, both will be counted. Regardless of precedence, if a

packet is to be dropped by a configured feature, it will be dropped. Ingress packets do not take precedence over egress packets nor do egress packets take precedence over ingress packets.

The precedence order from highest to lowest is as follows:

Meter Actions:

Port Access Client Policy
Port Policy
VLAN Policy
Ingress Global Policy

QoS Actions:

Port Access Client Policy Remark
Ingress/Egress Port Policy Remark
Ingress/Egress VLAN Policy Remark
Global Policy Remark
QoS DSCP Map Entry
QoS COS Map Entry
QoS Port Config
MAC Port ACL Logging
IP Port ACL Logging
MAC VLAN ACL Logging
IP VLAN ACL Logging

Redirect actions:

Policy L2 Tunnel
Port Access Client Policy Captive-portal

L4 port ranges

Any ACE or class entry that uses 'lt', 'gt', 'range', or port groups may use more than one hardware entry to represent the range of L4 ports.

Context group selectors

Context group selectors are a limited hardware resource that are required for applying ACLs and classifier policies. The selectors enable the application of an ACL or classifier policy to multiple instances of the same context (for example, ports on a line card or VLANs) without consuming additional resources.

There are a limited number of available context group selectors for each context group (Ingress Ports, Ingress VLANs, Egress Ports, Egress VLANs).



NOTE: IP ACLs require two selectors that are allocated together; one selector for each address family (IPv4 and IPv6).

Context group selectors work on a first-come-first-served basis. IP ACLs and Classes require two selectors that are allocated together; one selector for each address family (IPv4 and IPv6). Once all the group selectors for a context group have been used, no new application type of ACL or classifier policy for the context group can be applied.

Context group selector consumption and availability are as follows:

Type	Selectors
Ingress Port MAC ACL	1
Ingress Port IP ACL	2

Table Continued

Type	Selectors
Ingress Port Policy	1
Available Ingress Port Selectors	4
Ingress VLAN MAC ACL	1
Ingress VLAN IP ACL	2
Routed-Ingress VLAN IP ACL	2
Ingress VLAN Policy	1
Available Ingress VLAN Selectors	4
Egress Port MAC ACL	1
Egress Port IP ACL	2
Egress Port Policy	1
Available Egress Port Selectors	5
Egress VLAN MAC ACL	1
Egress VLAN IP ACL	2
Routed-Egress VLAN IP ACL	2
Egress VLAN Policy	1
Available Egress VLAN Selectors	5

ACL and Policy hardware resource commands

show resources

Syntax

```
show resources [<SLOT-ID>] [vsx-peer]
```

Description

Shows hardware resource consumption for the specified VSF member or for all VSF members. Resource data is updated every 10 seconds.

Hardware resource consumption information is shown for:

- TCAM entries
- TCAM lookups
- Policers

Command context

Operator (>) or Manager (#)

Parameters

<SLOT-ID>

Specifies the VSF member.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

The widths for show resources can have features combined (IPv4 + IPv6) into one TCAM lookup. Therefore, the table widths for each ACL/classifier policy type are variable depending on what is applied. For example:

```
"Ingress IP Port ACL" = Ingress v4 Port ACLs + Ingress v6 Port ACLs
                      = 1 TCAM entry + 4 TCAM entries
                      = 5 TCAM entries
```

Widths per feature are as follows:

```
MAC ACL      1
IPv4 ACL     1
IPv6 ACL     4
MAC Class    1
IPv4 Class   2
IPv6 Class   4
```

A MAC Class with an ethertype of "any" has a width of 7 because it uses one TCAM entry each for MAC, IPv4, and IPv6. Specifying the IPv4 (0x0800) or IPv6 (0x86DD) ethertypes in a MAC Class uses a TCAM entry equal to their respective size. IPv4 uses a width of 2 and IPv6 uses a width of 4.

Examples

Showing hardware resource consumption on a 6200 switch:

```
switch# show resources
```

Resource Usage:

Mod	Description	Resource	Used	Reserved	Free
1/1	Total				
		Ingress TCAM Entries	0	0	5120
		Egress TCAM Entries	0	0	2048
		Ingress Lookups	0		9
		Egress Lookups	0		4
		Ingress Policers	0		2047
		Egress Policers	0		2047

Accessing Aruba Support

Aruba Support Services	https://www.arubanetworks.com/support-services/
Aruba Support Portal	https://asp.arubanetworks.com/
North America telephone	1-800-943-4526 (US & Canada Toll-Free Number) +1-408-754-1200 (Primary - Toll Number) +1-650-385-6582 (Backup - Toll Number - Use only when all other numbers are not working)
International telephone	https://www.arubanetworks.com/support-services/contact-support/

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
Software licensing	https://lms.arubanetworks.com/
End-of-Life information	https://www.arubanetworks.com/support-services/end-of-life/
Aruba software and documentation	https://asp.arubanetworks.com/downloads

Accessing updates

To download product updates:

Aruba Support Portal

<https://asp.arubanetworks.com/downloads>

If you are unable to find your product in the Aruba Support Portal, you may need to search My Networking, where older networking products can be found:

My Networking

<https://www.hpe.com/networking/support>

To view and update your entitlements, and to link your contracts and warranties with your profile, go to the Hewlett Packard Enterprise Support Center **More Information on Access to Support Materials** page:

<https://support.hpe.com/portal/site/hpsc/aae/home/>



IMPORTANT: Access to some updates might require product entitlement when accessed through the Hewlett Packard Enterprise Support Center. You must have an HP Passport set up with relevant entitlements.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

<https://asp.arubanetworks.com/notifications/subscriptions> (requires an active Aruba Support Portal (ASP) account to manage subscriptions). Security notices are viewable without an ASP account.

Warranty information

To view warranty information for your product, go to <https://www.arubanetworks.com/support-services/product-warranties/>.

Regulatory information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at <https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

Additional regulatory information

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